Multigrid for the even-odd preconditioned Möbius domain-wall operator with a factorized Pauli–Villars coarse correction

Wei-Lun Chen
wlchenba@gmail.com

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Domain-wall fermions keep chiral symmetry up to effects that vanish as L_s grows, but inverting the Möbius domain-wall operator suffers from critical slowing down toward the physical light-quark mass. While preconditioning with the Pauli–Villars operator successfully moves the spectrum into the right half-plane to enable Krylov methods, previous attempts at a Pauli–Villars multigrid failed to outperform red-black CG due to three structural limitations identified by Boyle and Yamaguchi. Kanamori, Chen and Matsufuru showed that preconditioning with the Pauli–Villars operator and dividing out the site-diagonal part, $(D_{\mathrm{PV}}C_{\mathrm{PV}}^{-1})^{\dagger}DC^{-1}$, moves all eigenvalues into the right half-plane and compresses the high end of the spectrum.

In this work, we construct a two-grid method for its even-odd form in which the quark and Pauli–Villars factors are coarsened separately. Every aggregate contains the whole fifth dimension so that the coarse lattice is four-dimensional with nearest-neighbor couplings, and levels below the coarse lattice replace deflation. These choices address the three reasons Boyle and Yamaguchi gave for why this approach did not beat red-black CG. A single setup serves every value of Neff's α. We also prove that $\Gamma_5 D^{-1}\Gamma_5$ (with $\Gamma_5 = \gamma_5 R$ and R the reflection of the fifth dimension) acts as a robust preconditioner for the $D^{\dagger}$ solves required in HMC, which bypasses the need for an expensive second multigrid setup. The field of values of the preconditioned operator reaches into the left half-plane, yet its spectrum has $\mathrm{Re}\,\lambda \geq 0.70$, and its pseudospectra account for the convergence. On four quenched $24^3 \times 48$ configurations the iteration count grows by a factor of 1.6–4.1 from $m = 0.1$ to $m = 0.001$, against 18–27 for mixed-precision even-odd CGNE. At $m = 0.001$ a solve is 2.9–6.0 times faster than mixed-precision CGNE, and a point-source propagator takes half its time, setup included.

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I. Introduction

Many lattice computations need fermions whose chiral symmetry is broken only slightly at finite lattice spacing. Operators of different chirality then do not mix, which weak matrix elements require, as in the Standard Model calculation of $K \to \pi\pi$ decays and ε'/ε [1]. Precision calculations at the physical quark masses use such fermions as well: the hadronic vacuum polarization and light-by-light contributions to the muon $g-2$ [2, 3, 4], the $B \to \pi\ell\nu$ form factors on fine lattices with the heavy quark in the same Möbius domain-wall action [5], inclusive semileptonic B decays [6, 7, 8], and parton distribution amplitudes [9]. The overlap operator [10, 11] is exactly chiral on the lattice. Domain-wall fermions [12, 13, 14] cost less: they add a fifth dimension of extent L_s, in which a large negative Wilson mass $-M_5$ binds modes

of opposite chirality to its two boundaries, and the residual breaking falls as L_s grows. The Möbius generalization [15, 16, 17] adds two coefficients (b, c) that reduce the residual breaking at fixed L_s; $c = 0$ is Shamir's operator.

Solving with the domain-wall operator is expensive, because it is neither Hermitian nor positive and one application costs L_s four-dimensional Wilson hops. The standard solver is still the conjugate gradient method [20] on the normal equation of the even-odd operator (CGNE), whose condition number is the square of that of the operator. As the quark mass m decreases, the smallest eigenvalues of D approach m, the condition number of the normal equation grows like $1/m^2$ and the CG iterations like $1/m$ until the finite volume limits them: critical slowing down. For Wilson fermions, multigrid removed it. The low modes that slow down Krylov solvers are locally coherent [19]: a few of them, cut into blocks, span the low-mode space. Adaptive aggregation multigrid uses this, and its cost hardly depends on the quark mass [27, 28, 29, 30, 31, 32].

Multigrid for domain-wall fermions. For domain-wall fermions multigrid is harder. Because of the large negative Wilson mass, the spectrum encloses the origin. Non-Hermitian Krylov methods then converge only after a number of steps of the order of the system size [35, 36]. Boyle [36] lists four approaches that work to some extent. They differ in the operator that is coarsened.

- Cohen *et al.* [37] coarsened the squared operator, with blocks that cover the whole fifth dimension. Its coarse operator reaches two hops.
- HDCG [38] accelerates CGNE and coarsens the squared even-odd Schur complement. Its coarse stencil has 81 points for blocks of at least 4^4.
- HDCR [39] coarsens the Hermitian indefinite operator $\gamma_5 RD$ at $c = 0$, where R reflects the fifth dimension. Its coarse operator stays nearest-neighbor, which keeps the setup affordable in HMC.
- Brower, Clark, Howarth and Weinberg [41] noted that the Pauli–Villars preconditioned operator $D_{\mathrm{PV}}^{-1} D$ carries the physical four-dimensional degrees of freedom. They solved $D_{\mathrm{PV}}^{\dagger} D x = D_{\mathrm{PV}}^{\dagger} b$ on the whole lattice. Projecting the Wilson kernel slice by slice gives coarse operators $\hat{D}$ and $\hat{D}_{\mathrm{PV}}$, both nearest-neighbor, and the coarse correction solves $\hat{D}_{\mathrm{PV}}^{\dagger} \hat{D}\, \hat{e} = \hat{r}$. They tested Shamir fermions in the two-dimensional Schwinger model.

Boyle and Yamaguchi [40] compared these approaches for Shamir fermions and found that on large volumes HDCR was set up and solved faster than a single even-odd CG solve. They also carried the Pauli–Villars method of Ref. [41] to four-dimensional QCD. On a $16^3 \times 32$ Shamir configuration at $m = 0.001$ it needed about half the fine-matrix applications of red-black CG, but a solve took 1089–1904 s against 62 s (one right-hand side, setup excluded, double precision), and they named three reasons. The method works on the whole lattice, while red-black CG is the relevant baseline. Its coarse space keeps a fifth dimension of extent L_s, which makes the coarse solves costly. And its coarse solves are non-Hermitian and were not deflated, while the non-normal case would need the pseudospectrum and a singular value decomposition; they did not expect deflation alone to make the method competitive. Two further findings matter here. First, CG on the Schur complement $M_{ee} - M_{eo} M_{oo}^{-1} M_{oe}$ (e, o: the even and odd four-dimensional sites) needed fewer iterations than on the left-preconditioned $1 - M_{ee}^{-1} M_{eo} M_{oo}^{-1} M_{oe}$, 3224 against 3880. For the Möbius setups of this work the order is reversed (Appendix D). Second, an even-odd smoother did not work with the HDCR correction, which they attributed to eigenvectors that are not aligned.

Recent work by Boyle and collaborators. The multiple right-hand-side HDCG of Ref. [36] solves the normal equation of the even-odd operator with a two-level conjugate gradient. Its coarse space is spanned by blocked low modes from Chebyshev filters. Treating many right-hand sides at once makes the coarse space cheap, and gains of up to twenty over even-odd CG are reported at physical quark mass. In a Lattice 2026 talk, Boyle and Oare [43] named two limits of this approach: forming the 81-point coarse

operator is expensive, and a single solve on a new gauge field, as in HMC, gains nothing from many right-hand sides. They coarsen the two-hop operator $D_{\mathrm{PV}}^{\dagger}D$ instead, which gives a 33-point stencil. They stressed that $D_{\mathrm{PV}}^{\dagger}D$ is non-normal: its field of values can reach the left half-plane although its spectrum does not, so the GCR bound of Eisenstat, Elman and Schultz [44] is lost. They added a shift to the smoothers. On a $48^3 \times 96$ lattice at the physical quark mass they reported 578 fine-matrix applications, against 1100 for HDCG and 26500 for red-black CG. The coarse solve took 90% of the time, and they named the missing Lanczos deflation of the coarse operator as its limit.

Setup cost in ensemble generation. A multigrid setup is paid once per gauge field. This is affordable for propagators but not for HMC, where the field changes at every step. Eick and Mawhinney [54] avoid the setup by renormalization-group blocking: a fine Möbius ensemble is blocked to a coarse lattice whose domain-wall operator approximates the low modes of the fine one. They find that the interpolated correction carries high modes that the fine operator amplifies by four orders of magnitude, and they need a Chebyshev low-pass filter of degree 150–300.

The preconditioned operator. Kanamori, Chen and Matsufuru [45] examined the spectra of several preconditioned Möbius operators, looking for one local enough for a multigrid. Since the Pauli–Villars mass is of the cutoff scale, they tried its block-diagonal part B_{PV}, whose blocks would be the coarse sites, as an approximation of D_{PV}. With $B_{\mathrm{PV}}^{\dagger}$, however, and also with B_{PV}^{-1} applied approximately block by block, eigenvalues with negative real part reappear, so the Pauli–Villars factor has to be kept whole. Dividing out the site-diagonal part C in addition lowers the high end of the spectrum. The resulting operator $A' = (D_{\mathrm{PV}}C_{\mathrm{PV}}^{-1})^{\dagger}DC^{-1}$ has all eigenvalues in the right half-plane and the smallest condition number among those they studied. Separately, Neff's form of the domain-wall matrix [46, 47] rescales the fifth dimension by a site-local factor Q and changes the conditioning of the even-odd system.

This work. We construct a two-grid method for A' that takes up the Pauli–Villars route of Refs. [41, 40] and addresses the three reasons of Ref. [40] in turn. It is an adaptive aggregation multigrid in the sense of Refs. [27, 30, 32]: the coarse space comes from computed near-null vectors cut into aggregates, and the coarse operators are Galerkin projections. First, the outer system is the even-odd one, as for red-black CG. Its inverse contains the inverses of the quark operator and the adjoint Pauli–Villars operator, with site-local factors and the restriction to the even sites between them, and we replace each inverse by a Galerkin inverse, as Ref. [41] does on the whole lattice. Second, every aggregate contains the whole fifth dimension, so the coarse lattice is four-dimensional and both coarse operators are nearest-neighbor. Third, the Pauli–Villars coarse system is solved in a Σ-Hermitian form by MINRES, and levels below the coarse lattice with a dense LU at the bottom take the place of deflation (Sec. III.B). The test vectors are approximate right singular vectors of the non-normal operator. The prolongation lifts these vectors to the whole lattice in two ways, one for each coarse solve; for the right form this makes the correction exact on them (Eq. (3.8)). The left form used in practice is not exact, but with the same vectors y it still needs $\hat{t}$ (Sec. V.C.2).The smoother is a fixed Chebyshev polynomial, and the setup combines Chebyshev filtering with a bootstrap. For HMC we also solve with $D^{\dagger}$. We show that the multigrid of D, placed between two $\Gamma_5 = \gamma_5 R$, is a good preconditioner for $D^{\dagger}$, with no second setup. We test the method on quenched configurations from 12^4 to $24^3 \times 48$ on a single GPU against even-odd CGNE and single-grid Krylov solvers on $\tilde{A}_L$; a comparison with HDCG and with the coarsening of $D_{\mathrm{PV}}^{\dagger}D$ in one implementation is left to future work.

Section II recalls the spectra the construction rests on and defines the operator, its preconditioning and the even-odd forms. Section III gives the two-grid method: the factorized correction, the coarse solvers, the setup, the smoother and the outer solver. Section IV solves with $D^{\dagger}$. Section V gives the results: first the cost against CGNE and with $D^{\dagger}$, then what each part of the method contributes, then why it converges. Section VI sums up. Section VII lists open problems. Appendix A relates our notation to other references, Appendix B shows the spectra of other preconditioners, Appendix C lists the gauge

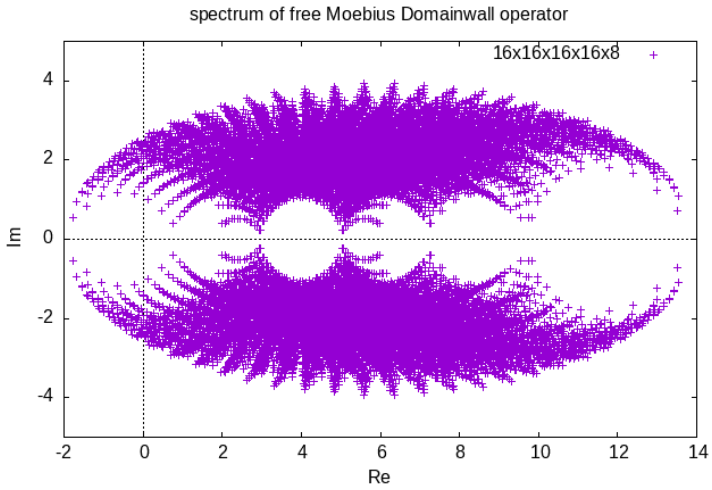


FIG. 1. Spectrum of a free Möbius domain-wall operator on $16^4 \times 8$ with quark mass 0.01, $b = 1.5$, $c = 0.5$ and domain-wall height $M_5 = 1$. From Ref. [45], Fig. 1.

configurations, Appendix D describes the reference CG.

II. The operator and its preconditioning

A. Spectra of preconditioned Möbius operators

Where the spectrum lies decides which iterative methods can work. A Krylov method builds a residual polynomial p with $p(0) = 1$ that should be small on the spectrum. Such a polynomial of low degree exists only if a region that excludes the origin, such as an ellipse, encloses the spectrum. The Chebyshev smoother of Sec. III.D needs the same, since it is defined by an ellipse in the right half-plane. Both therefore need the eigenvalues on one side of the origin. Those of the domain-wall operator do not, and the preconditioners below move them there. Figures 1 and 2, from Ref. [45], show the spectrum without and with both preconditioners; Fig. 7 in Appendix B shows the other preconditioners discussed below.

The unpreconditioned operator. The free spectrum (Fig. 1) is a band that reaches into $\mathrm{Re}\,\lambda < 0$. On a $2 + 1$-flavor configuration at the physical point (Appendix B, Fig. 7) the real parts lie in $[-1.29, 13.2]$. So CG and BiCGStab [21] do not work on D itself, and the practical choice has been CG on a positive definite squared operator.

Site-diagonal preconditioning. The site-diagonal part C has an inverse that is the same matrix on every site (Sec. II.B.3). With it, DC^{-1} has real parts in $[-0.22, 2.22]$. The spectrum shrinks, but it still lies on both sides of the origin.

Pauli–Villars preconditioning. The operator $D_{\mathrm{PV}}^{-1}D$, with $D_{\mathrm{PV}} = D(m = 1)$, approximates the overlap operator [41]. Since D_{PV}^{-1} costs as much as a solve with D, Ref. [41] used $D_{\mathrm{PV}}^{\dagger}$ instead. $D_{\mathrm{PV}}^{\dagger}D$ has positive real parts. Its low modes lie close to the imaginary axis, shifted by about the quark mass, as for an overlap operator. Its high end, however, reaches about 187, which an overlap operator would not. If we replace D_{PV} by its block-diagonal approximation B_{PV}, which drops the hops between 4^4 blocks, negative real parts come back.

Both together. The operator $A' = (D_{\mathrm{PV}}C_{\mathrm{PV}}^{-1})^{\dagger}DC^{-1}$ has real parts in $[2.67 \times 10^{-4},\ 5.22]$ (Fig. 2, left). Its condition number, 1.96×10^4, is smaller than the 2.34×10^4 of $D_{\mathrm{PV}}^{\dagger}D$ and the smallest in Ref. [45]. With D_{PV} replaced by B_{PV}, the real parts become negative again (right). So a preconditioner that truncates the Pauli–Villars factor to blocks would inherit eigenvalues with negative real part. Our method

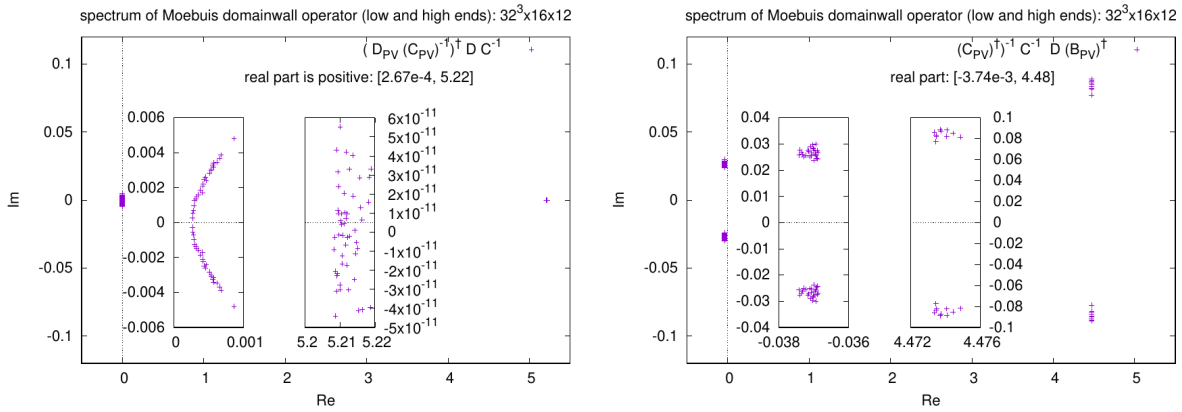


FIG. 2. Left: spectrum of $A' = (D_{\rm PV}C_{\rm PV}^{-1})^\dagger D C^{-1}$, Eq. (2.18), on a JLQCD $2+1$-flavor configuration at the physical point; all real parts are positive. Right: the same with $D_{\rm PV}$ replaced by its block-diagonal approximation $B_{\rm PV}$; negative real parts reappear. The text of Ref. [45] quotes $48^3 \times 16$ with $L_s = 12$; the panel titles read $32^3 \times 16 \times 12$. From Ref. [45], Fig. 4.

keeps the factor exact on the fine lattice and approximates it only through its Galerkin coarse operator (Sec. III.A).

Two caveats. First, the figures show the full-lattice operator A', not the even-odd operators of Sec. II.C. Second, positive real parts of the eigenvalues do not mean a positive field of values. For a non-normal operator the GCR bound [44] needs the Hermitian part $(A + A^\dagger)/2$ to be positive definite. This can fail while every eigenvalue lies in the right half-plane, as Ref. [43] discusses for $D_{\rm PV}^\dagger D$. It occurs for the even-odd form: for $\tilde{A}_R$ itself the field of values reaches just below zero on 12^4 ($\mathrm{Re}\, W \geq -4.6\times10^{-3}$ with stout smearing, -5.0×10^{-2} without), and for the multigrid-preconditioned $\tilde{A}_R\mathcal{M}$ far below (Sec. V.D.1). The bound of Ref. [44] is then lost, but the spectrum of $\tilde{A}_R\mathcal{M}$ stays at $\mathrm{Re}\,\lambda \geq 0.70$ and its pseudospectra bound full GMRES (Secs. V.D.2, V.D.3); truncated GCR is not covered (Sec. V.D.4).

B. The operator and Pauli–Villars preconditioning

This subsection defines the Möbius operator in the form of Brower, Neff and Orginos [17] and derives from it the quantities the method is built on: the part C of the operator that acts within a four-dimensional site and its inverse, the Γ_5-transformation of D, the Pauli–Villars preconditioned operator A' of Ref. [45], and Neff's α. The notation follows Ref. [17] except where stated; Appendix A relates it to Ref. [45] and to Bridge++ symbol by symbol.

1. The Möbius operator

Ingredients. ψ is a field on the four-dimensional lattice with an extra index $s = 0, \ldots, L_s - 1$, the fifth dimension; ψ_s is an ordinary Dirac field for each s. $P_\pm = \frac{1}{2}(1 \pm \gamma_5)$ are the chiral projectors. The Wilson hopping term is

$$(K\psi)(x) = \sum_{\mu=1}^{4} \Big[(1-\gamma_\mu) U_\mu(x) \psi(x+\hat{\mu}) + (1+\gamma_\mu) U_\mu^\dagger(x-\hat{\mu}) \psi(x-\hat{\mu}) \Big], \tag{2.1}$$

and the Wilson operator with mass $-M_5$ (the domain-wall height $M_5 > 0$) is

$$W = (4 - M_5) - \tfrac{1}{2} K. \tag{2.2}$$

This is the usual $D_W(-M_5)$ with hopping coefficient $\frac{1}{2}$; in the κ normalization $W = (4-M_5)(1-\kappa K)$ with $\kappa = 1/(8-2M_5)$. W acts on each slice s in the same way and does not mix slices. Its free spectrum, needed in Sec. IV, is $\{-M_5 + \sum_\mu (1-\cos p_\mu) \pm i(\sum_\mu \sin^2 p_\mu)^{1/2}\}$.

The operator. With

$$D_+ = 1 + bW, \qquad D_- = cW - 1 \tag{2.3}$$

(Ref. [17], Eq. (2.18)), the Möbius operator at quark mass m acts slice by slice as

$$(D\psi)_s = D_+\psi_s + D_-\big(P_-\psi_{s+1} + P_+\psi_{s-1}\big), \qquad \psi_{L_s} \equiv -m\,\psi_0, \quad \psi_{-1} \equiv -m\,\psi_{L_s-1} \tag{2.4}$$

(Ref. [17], Eq. (2.20), with s counted from 0). For $L_s = 4$, as a matrix in s,

$$D = \begin{pmatrix} D_+ & D_-P_- & 0 & -mD_-P_+ \\ D_-P_+ & D_+ & D_-P_- & 0 \\ 0 & D_-P_+ & D_+ & D_-P_- \\ -mD_-P_- & 0 & D_-P_+ & D_+ \end{pmatrix}. \tag{2.5}$$

Shamir's operator is $b = 1$, $c = 0$; the results of this work use $b = 1.5$, $c = 0.5$ ($b - c = 1$, $b + c = 2$). The quark mass enters only through the two corners. The physical four-dimensional quark field is $q = P_-\psi_0 + P_+\psi_{L_s-1}$.

The hop in the fifth dimension. Eq. (2.4) separates into a part that hops in the fifth dimension and one that does not. Define

$$(Y\psi)_s = -P_-\psi_{s+1} - P_+\psi_{s-1} \qquad \text{(with the same wrap-around)}, \tag{2.6}$$

the off-diagonal part of Eq. (2.5) without D_- and with a minus sign. Then $D = D_+ - D_-Y$. Here D_- stands to the *left* of Y, since in Eq. (2.4) it acts on the shifted and projected field, and this order matters in Sec. II.B.2. Inserting Eqs. (2.2) and (2.3),

$$D = 1 + bW - (cW - 1)Y = 1 + Y + W\,(b - cY) = C - \tfrac{1}{2}K\,(b - cY), \qquad C = d_+ + d_-Y, \tag{2.7}$$

with $d_+ = 1 + b(4 - M_5)$ and $d_- = 1 - c(4 - M_5)$. In the last step $W = (4 - M_5) - \frac{1}{2}K$ was split into its site-local constant and its hop; the constant parts form C. Note $d_- = -(c(4 - M_5) - 1)$ is *minus* the site-local part of D_-, because Y carries the sign. The Pauli–Villars operator is the same operator at unit mass, $D_{\mathrm{PV}} = D|_{m=1}$, $C_{\mathrm{PV}} = C|_{m=1}$; $D_\pm$ and K do not depend on m.

2. Γ_5-transformation and the exact adjoint

For the Wilson operator $\gamma_5 W \gamma_5 = W^\dagger$, so a solve with $W^\dagger$ is a solve with W between two multiplications by γ_5. The analogue for domain-wall fermions uses

$$\Gamma_5 = \gamma_5 R, \qquad (R\psi)_s = \psi_{L_s-1-s}. \tag{2.8}$$

Each building block transforms as follows.

1. W, D_+, D_-: they act on every slice alike, so they commute with R, and $\gamma_5 W \gamma_5 = W^\dagger$. Hence $\Gamma_5 W \Gamma_5 = W^\dagger$, $\Gamma_5 D_\pm \Gamma_5 = D_\pm^\dagger$.

2. Y: it has no hop in the four dimensions, so $\Gamma_5 Y \Gamma_5$ can be read off Eq. (2.6). γ_5 commutes with $P_\pm$; R turns the hop $s \to s+1$ into $s \to s-1$. So $\Gamma_5 Y \Gamma_5$ has $-P_-$ on the hop down and $-P_+$ on the hop up, and the corner $-m$ moves to the other corner. This is the transpose of Eq. (2.6) in s with $P_\pm$ in place, which is $Y^\dagger$ since $P_\pm^\dagger = P_\pm$ and m is real: $\Gamma_5 Y \Gamma_5 = Y^\dagger$.

3. W and Y do not commute: γ_μ in K anticommutes with γ_5, so $\gamma_\mu P_\pm = P_\mp \gamma_\mu$, and the hop of K exchanges the two projectors of Y.

From $D = D_+ - D_-Y$ it follows that

$$\Gamma_5 D \Gamma_5 = D_+^\dagger - D_-^\dagger Y^\dagger, \qquad \text{whereas} \qquad D^\dagger = D_+^\dagger - Y^\dagger D_-^\dagger, \tag{2.9}$$

the adjoint reversing the order of the product. The difference is a commutator,

$$\Gamma_5 D \Gamma_5 - D^\dagger = Y^\dagger D_-^\dagger - D_-^\dagger Y^\dagger = c\,[Y^\dagger, W^\dagger] = \tfrac{c}{2}\big[K^\dagger, Y^\dagger\big], \tag{2.10}$$

where $D_-^\dagger = cW^\dagger - 1$ and $W^\dagger = (4 - M_5) - \frac{1}{2}K^\dagger$ were used. It vanishes for Shamir's operator ($c = 0$) and is of order c otherwise. Thus $\Gamma_5 D \Gamma_5 \neq D^\dagger$ for Möbius fermions: a multigrid built for D cannot be used for $D^\dagger$ by conjugating with Γ_5. On a 12^4 configuration with stout links ($M_5 = 1$, $m = 0.001$), $\|\Gamma_5 D \Gamma_5 b - D^\dagger b\|/\|D^\dagger b\| = 0.15$ for $(b, c) = (1.5, 0.5)$, 0.22 for $(2, 1)$, and 0 for $(1, 0)$, for a random b.

The exact relation. D_+ and D_- are functions of the same W and commute. Where D_- is invertible, factor it out to the left:

$$D = D_-\, K_D, \qquad K_D = D_-^{-1} D_+ - Y. \tag{2.11}$$

K_D is Γ_5-Hermitian: by item 1, $\Gamma_5 D_-^{-1} D_+ \Gamma_5 = (D_-^\dagger)^{-1} D_+^\dagger$, and this equals $(D_-^{-1} D_+)^\dagger = D_+^\dagger (D_-^\dagger)^{-1}$ because $D_+^\dagger$ and $D_-^\dagger$ commute; with item 2, $\Gamma_5 K_D \Gamma_5 = K_D^\dagger$. Then

$$D^\dagger = K_D^\dagger D_-^\dagger = \Gamma_5 K_D \Gamma_5\, D_-^\dagger = \Gamma_5 D_-^{-1} D\, \Gamma_5 D_-^\dagger, \tag{2.12}$$

and multiplying by $D_-\Gamma_5$ from the left,

$$\boxed{D_-\, \Gamma_5\, D^\dagger = D\, \Gamma_5\, D_-^\dagger} \tag{2.13}$$

This form has no inverse, holds for any m (so also for $D_{\rm PV}$, with the same D_-), and can be checked by applying both sides to a vector. On the configuration above both sides agree to 3×10^{-16} for $(b, c) = (1.5, 0.5)$ and $(2, 1)$. The Hermitian operator of the Möbius case is $\Gamma_5 K_D = \Gamma_5 D_-^{-1} D$; Ref. [17], Eq. (2.22), defines its Γ_5 as $\gamma_5 R D_-^{-1}$ for that reason. It contains D_-^{-1}, whose cost is discussed in Sec. IV.

3. The site-diagonal part and its inverse

$C = d_+ + d_- Y$ of Eq. (2.7) is what remains of D when the four-dimensional hops are dropped. It acts within one four-dimensional site, as an $L_s \times L_s$ matrix in s times the identity in color, and it commutes with $P_\pm$ (only Y is involved, and Y maps $P_+\psi$ to $P_+\psi$ and $P_-\psi$ to $P_-\psi$). So it splits into a block on $\psi^+ = P_+\psi$ and one on $\psi^- = P_-\psi$. From Eq. (2.6), Y moves ψ^+ from $s-1$ to s and ψ^- from $s+1$ to s, with the factor -1, and the wrap-around gives $+m$:

$$(C\psi)_s^+ = \begin{cases} d_+ \psi_0^+ + m\, d_- \psi_{L_s-1}^+, & s = 0, \\ d_+ \psi_s^+ - d_- \psi_{s-1}^+, & s \geq 1, \end{cases} \tag{2.14}$$

a lower bidiagonal matrix with one corner element; for ψ^- the same with $s \to L_s - 1 - s$ (upper bidiagonal). For $L_s = 4$ the block on ψ^+ is

$$C^+ = \begin{pmatrix} d_+ & 0 & 0 & m\, d_- \\ -d_- & d_+ & 0 & 0 \\ 0 & -d_- & d_+ & 0 \\ 0 & 0 & -d_- & d_+ \end{pmatrix}. \tag{2.15}$$

The inverse. To solve $C^+x = r$, write the rows $s \geq 1$ as $x_s = (r_s + d_- x_{s-1})/d_+$. Starting from an unknown x_0, each x_s is affine in x_0: with $q = d_-/d_+$,

$$x_s = p_s + q^s x_0, \qquad p_0 = 0, \quad p_s = (r_s + d_- p_{s-1})/d_+, \tag{2.16}$$

which is checked by induction: $(r_s + d_-(p_{s-1} + q^{s-1}x_0))/d_+ = p_s + q^s x_0$. The row $s = 0$, $d_+ x_0 + m\, d_- x_{L_s-1} = r_0$, with $x_{L_s-1} = p_{L_s-1} + q^{L_s-1}x_0$, fixes x_0:

$$x_0 = \frac{r_0 - m\, d_-\, p_{L_s-1}}{d_+ + m\, d_-\, q^{L_s-1}}, \qquad x_s = p_s + q^s x_0. \tag{2.17}$$

For C^- the substitution runs from $s = L_s - 1$ downwards. $C^\dagger$ has the transposed blocks: on ψ^+ an upper bidiagonal matrix, solved like C^-, and on ψ^- a lower one. C^{-1} is thus exact, costs $O(L_s)$ per site, and does not depend on the gauge field [45]. It exists unless $d_+ + m\, d_- q^{L_s-1} = 0$.

4. The preconditioned operator

The spectrum of D surrounds the origin (Sec. II.A), so Krylov methods and polynomial smoothers cannot work on it directly. Ref. [45] multiplies $D\psi = \eta$ from the left by $(D_{\rm PV} C_{\rm PV}^{-1})^\dagger$ and substitutes $\psi = C^{-1}y$:

$$A'y = (D_{\rm PV} C_{\rm PV}^{-1})^\dagger\, \eta, \qquad A' = (D_{\rm PV} C_{\rm PV}^{-1})^\dagger\, D\, C^{-1}, \qquad \psi = C^{-1}y. \tag{2.18}$$

The two factors are chosen for these reasons:

- DC^{-1} is D with its site-local part divided out; its eigenvalues are much more compact than those of D (Sec. II.A), but still on both sides of the origin.
- For $m = 1$, $DC^{-1} = D_{\rm PV} C_{\rm PV}^{-1}$ and $A' = (D_{\rm PV} C_{\rm PV}^{-1})^\dagger (D_{\rm PV} C_{\rm PV}^{-1})$, which is Hermitian and positive definite. For $m < 1$ the two factors differ only in the corners of Eq. (2.5), and A' is a perturbation of a positive operator.
- $D_{\rm PV}^{-1} D$ approximates the overlap operator [41]; $D_{\rm PV}^\dagger$ in place of $D_{\rm PV}^{-1}$ keeps the cost of one application of A' at two applications of D plus the site-local C^{-1} and $C_{\rm PV}^{-1}$.

5. Link smearing

The fermion operator may be built from smeared links in place of $U_\mu(x)$; smearing integrates the surrounding gauge field into each link and suppresses the short-distance fluctuations that produce near-zero modes of W. We use the stout projection combined with the APE staples [18]:

$$C_\mu(x) = \sum_{\nu \neq \mu} \rho \Big[U_\nu(x) U_\mu(x+\hat\nu) U_\nu^\dagger(x+\hat\mu) + U_\nu^\dagger(x-\hat\nu) U_\mu(x-\hat\nu) U_\nu(x-\hat\nu+\hat\mu) \Big], \tag{2.19}$$

$$U'_\mu(x) = \exp\Big(\big[C_\mu(x) U_\mu^\dagger(x) \big]_{\rm AT} \Big)\, U_\mu(x), \tag{2.20}$$

where $[\cdots]_{\rm AT}$ is the anti-Hermitian traceless part and ρ a tunable parameter. The step (2.20) is applied repeatedly; "stout $\rho \times n$" means n steps with the same ρ. Only W (through K) sees the smeared links; Y, C and Q do not depend on the gauge field. In HMC the force of the fermion action is taken with respect to the smeared links and carried back to U through the derivative of Eq. (2.20).

6. Neff's α

Neff's form of the domain-wall matrix [46, 47] is $D^{(\alpha)} = DQ$ with a site-local Q, diagonal in the fifth dimension,

$$Q_0 = P_- + \alpha P_+, \qquad Q_s = \alpha \;\; (0 < s < L_s - 1), \qquad Q_{L_s-1} = P_+ + \alpha P_-. \tag{2.21}$$

The following properties, each read off Eq. (2.21), are used below:

- Q is Hermitian, and Q^{-1} is Eq. (2.21) with $\alpha \to 1/\alpha$.
- Q leaves the components $P_-\psi_0$ and $P_+\psi_{L_s-1}$ unchanged, which make up the quark field q; so $D^{(\alpha)}$ and D have the same four-dimensional propagator.
- The solution of $D\psi = \eta$ is $\psi = Q\psi^{(\alpha)}$ with $D^{(\alpha)}\psi^{(\alpha)} = \eta$, since $D^{(\alpha)}\psi^{(\alpha)} = DQ\psi^{(\alpha)} = D\psi$.
- Q is site-local, so the site-diagonal part of DQ is CQ, with $(CQ)^{-1} = Q^{-1}C^{-1}$ and $(CQ)^\dagger = QC^\dagger$.

The factor may be applied to the quark operator alone or to the Pauli–Villars operator as well; we write $D_{\mathrm{PV}}^{(\alpha)} = D_{\mathrm{PV}}Q$ in the latter case. For $\alpha < 1$ it improves the conditioning of the even-odd operator [46, 47].

C. Even-odd forms

The diagonal blocks are C. Order the sites so that the even ones come first. We write X_{pq} for the block of an operator X that maps the sites of parity q to those of parity p, and ψ_e, ψ_o for the even and odd parts of a field. The hopping term K connects nearest neighbors, which always have opposite parity, so it has no even–even or odd–odd block. In $D = C - \frac{1}{2}K(b - cY)$ (Eq. (2.7)) the diagonal blocks therefore come from C alone and the off-diagonal ones from the hop alone:

$$D = \begin{pmatrix} D_{ee} & D_{eo} \\ D_{oe} & D_{oo} \end{pmatrix} = \begin{pmatrix} C_e & D_{eo} \\ D_{oe} & C_o \end{pmatrix}, \qquad D_{pq} = -\tfrac{1}{2}K_{pq}(b - cY) \ \ (p \neq q), \qquad C = \begin{pmatrix} C_e & 0 \\ 0 & C_o \end{pmatrix}. \tag{2.22}$$

So the D_{ee} and D_{oo} of the even-odd literature (Ref. [47], Eq. (18); Ref. [45], Eq. (4)) *are* the site-diagonal part: $D_{ee} = C_e$ and $D_{oo} = C_o$ are the blocks of the same C that appears in $A' = (D_{\mathrm{PV}}C_{\mathrm{PV}}^{-1})^\dagger DC^{-1}$, restricted to the even and to the odd sites. Since C is the same $L_s \times L_s$ matrix on every site (Sec. II.B.3), C_e and C_o are also the same matrix, applied on different sites; the subscript only says where the field lives. We keep the letter C for them to make this visible, and the letter D for the off-diagonal blocks, which carry the gauge field. The same holds for D_{PV} with $C_{\mathrm{PV},e}$, $C_{\mathrm{PV},o}$.

With these blocks the Schur complement on the even sites is $S = C_e - D_{eo}C_o^{-1}D_{oe}$, and S_{PV} is that of D_{PV}. The injection ι puts a field on the even sites onto the whole lattice, with zero on the odd sites, and the projection $\pi = \iota^\dagger$ takes the even sites:

$$\iota y = \begin{pmatrix} y \\ 0 \end{pmatrix}, \qquad \pi \begin{pmatrix} \psi_e \\ \psi_o \end{pmatrix} = \psi_e, \qquad \iota\pi \begin{pmatrix} \psi_e \\ \psi_o \end{pmatrix} = \begin{pmatrix} \psi_e \\ 0 \end{pmatrix}, \tag{2.23}$$

So $\iota\pi$ sets the odd sites to zero and leaves a field that already vanishes there unchanged. Two identities are used throughout, $S^{-1} = \pi D^{-1}\iota$ and $(S_{\mathrm{PV}}^\dagger)^{-1} = \pi(D_{\mathrm{PV}}^\dagger)^{-1}\iota$. To see them, solve $D\psi = \iota\eta_e = (\eta_e, 0)^{\mathsf{T}}$. The odd rows give $\psi_o = -C_o^{-1}D_{oe}\psi_e$, and the even rows then read $S\psi_e = \eta_e$. Hence $S^{-1}\eta_e = \pi D^{-1}\iota\eta_e$, and the same argument for $D_{\mathrm{PV}}^\dagger$ gives the second identity.

Two forms. The site-diagonal factor can stand on either side of S. This gives the right and the left form,

$$\tilde{A}_R = (S_{\mathrm{PV}}C_{\mathrm{PV},e}^{-1})^\dagger\,(S\,C_e^{-1}), \qquad \tilde{A}_L = (C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^\dagger\,(C_e^{-1}S). \tag{2.24}$$

$\tilde{A}_R$ is the even-odd counterpart of A'. $C_e^{-1}S$ is the usual left-preconditioned even-odd operator, which Boyle and Yamaguchi found worse than S itself for CG on the normal equation [40].

With Neff's α. Replace D by DQ. Since Q acts within a site and is diagonal in s, the blocks of DQ are C_eQ_e, $D_{eo}Q_o$, $D_{oe}Q_e$ and C_oQ_o. Its Schur complement is therefore

$$C_eQ_e - D_{eo}Q_o(C_oQ_o)^{-1}D_{oe}Q_e = SQ_e, \tag{2.25}$$

and its diagonal block is $C_e Q_e$. In the right form the two factors of Q_e cancel: $SQ_e(C_eQ_e)^{-1} = SC_e^{-1}$. So $\tilde{A}_R$ does not depend on α. In the left form they do not cancel. With $(C_eQ_e)^{-1}SQ_e = Q_e^{-1}C_e^{-1}SQ_e$ and Q Hermitian,

$$\tilde{A}_L^{(\alpha)} = \begin{cases} Q_e \left(C_{\mathrm{PV},e}^{-1} S_{\mathrm{PV}}\right)^\dagger Q_e^{-2} \left(C_e^{-1} S\right) Q_e, & \alpha \text{ in } D \text{ and } D_{\mathrm{PV}}, \\ \left(C_{\mathrm{PV},e}^{-1} S_{\mathrm{PV}}\right)^\dagger Q_e^{-1} \left(C_e^{-1} S\right) Q_e, & \alpha \text{ in } D \text{ only}. \end{cases} \tag{2.26}$$

For $D\psi = \eta$, with $\eta'_e = \eta_e - D_{eo}C_o^{-1}\eta_o$, the right-hand side is $\rho_R = (S_{\mathrm{PV}}C_{\mathrm{PV},e}^{-1})^\dagger \eta'_e$ or $\rho_L = (C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^\dagger C_e^{-1}\eta'_e$ (operators with Q when $\alpha \neq 1$); the solution is $\psi_e^{(\alpha)} = C_e^{-1}y$ (right) or y (left), $\psi_o^{(\alpha)} = C_o^{-1}(\eta_o - D_{oe}\psi_e^{(\alpha)})$, $\psi = Q\psi^{(\alpha)}$.

Why both forms. The two forms serve different purposes. The factorized correction of Sec. III.A is exact for the right form (Eq. (3.8)), but Q cancels in it, so the right form cannot use Neff's α. The left form keeps α. The correction is not exact for it, yet it converges within a few iterations of the right form, and with α it can save iterations. The outer solver therefore uses the left form in practice. The measurements of Sec. V are on the left form unless stated.

How α acts on CG and on the multigrid. The left Schur factor of DQ is a similarity transform of that of D:

$$\mathcal{K}^{(\alpha)} = (C_eQ_e)^{-1}SQ_e = Q_e^{-1}\mathcal{K}Q_e, \qquad \mathcal{K} = C_e^{-1}S. \tag{2.27}$$

So $\mathcal{K}^{(\alpha)}$ and $\mathcal{K}$ have the same eigenvalues. Their singular values differ, because Q_e is not unitary: its entries are α in the bulk and 1 on half of the wall components. CG on the normal equation converges at a rate set by the ratio of the extreme singular values, so it sees the change at every mass. The multigrid solves $\tilde{A}_L^{(\alpha)}$, Eq. (2.26), where the Pauli–Villars factor stands outside the two Q_e; α changes the eigenvalues of that operator too, but the iterations of the multigrid follow only the part of the spectrum that the smoother damps, since the coarse correction removes the low modes for any α. We therefore expect α to help the multigrid less than CG, and least where the coarse correction sets the iterations. At $m = 0.001$ on $24^3 \times 48$, $\alpha = 0.6$ saves CG 25–35% of its iterations and the multigrid at most two (Sec. V.B.1).

Eq. (2.27) also lets CG use α in HMC without changing the action. The even-odd action needs $(\mathcal{K}^\dagger\mathcal{K})^{-1}$, and with $\mathcal{K} = Q_e\mathcal{K}^{(\alpha)}Q_e^{-1}$,

$$(\mathcal{K}^\dagger\mathcal{K})^{-1} = Q_e\left[(\mathcal{K}Q_e)^\dagger(\mathcal{K}Q_e)\right]^{-1}Q_e, \tag{2.28}$$

a normal equation of $\mathcal{K}Q_e = Q_e\mathcal{K}^{(\alpha)}$, whose singular values are close to those of $\alpha\mathcal{K}^{(\alpha)}$ (they differ only by the wall components of Q_e).

III. The two-grid method

A multigrid preconditioner removes the error of an iterate in two parts. A few steps of a cheap iteration on the fine lattice, the *smoother*, damp the high modes. The low modes, which make CG slow, are removed by solving a smaller problem on a coarse lattice. The fine lattice is cut into blocks, here of 4^4 sites with the whole fifth dimension; each block becomes one coarse site, whose degrees of freedom are a few approximate low modes restricted to the block, the *test vectors*. The prolongation $\mathcal{P}$ maps a coarse vector to the fine lattice through them, and the coarse operator of a fine operator X is its Galerkin projection $\mathcal{P}^\dagger X\mathcal{P}$, which couples only neighboring coarse sites if X couples only neighboring fine sites. A few vectors per block suffice because low modes are locally coherent [19]. Finding them, the *setup*, is paid once per gauge field. The setup uses only the plain D and D_{PV}; Neff's α enters only the outer operator $\tilde{A}_L^{(\alpha)}$, the smoother that works on it, and the site-local factors of the correction (3.2), so one setup serves both forms and every α.

TABLE 1. Multigrid methods for domain-wall fermions: the system solved, the operator that is coarsened, and its coarse stencil, for blocks that contain the whole fifth dimension (except MG-PV, whose coarse lattice keeps the fifth dimension) and are at least two sites wide in each direction (4^4 for HDCG). S is the Schur complement on the even sites.

method	system solved	operator coarsened	coarse stencil
HDCG [38, 36]	even sites, $S^\dagger S$	$S^\dagger S$ (four hops)	81 points
HDCR [39]	whole lattice, $\gamma_5 RD$	$\gamma_5 RD$, $c = 0$ (one hop)	9 points
MG-PV [41, 40]	whole lattice, $D_{\rm PV}^\dagger D$	D and $D_{\rm PV}$ apart, $c = 0$	9 points per slice
Boyle and Oare [43]	whole lattice, $D_{\rm PV}^\dagger D$	$D_{\rm PV}^\dagger D$ (two hops)	33 points
this work	even sites, $\tilde{A}_R$ or $\tilde{A}_L$	D and $D_{\rm PV}$ apart (one hop each)	9 points each

A. The factorized coarse-grid correction and the prolongation

1. Why factorize

Solving on the even sites halves the problem and improves the conditioning, and red-black CG, the solver to beat, works there. But the Schur complement reaches two hops, so the existing methods make a choice (Table 1). HDCG solves on the even sites and coarsens the squared Schur complement, with an 81-point coarse stencil [38]. The methods with a small coarse stencil work on the whole lattice: HDCR [39] coarsens $\gamma_5 RD$, Boyle and Oare [43] coarsen $D_{\rm PV}^\dagger D$, and MG-PV [41, 40] coarsens D and $D_{\rm PV}$ separately on a coarse lattice that keeps the fifth dimension. Two of the three reasons of Ref. [40] are these choices of MG-PV: the whole lattice, and the fifth dimension in the coarse space. We keep the separate coarse operators of D and $D_{\rm PV}$, use them for the even-odd operator, and make the coarse lattice four-dimensional. The third reason, the coarse solves, is taken up in Sec. III.B.

The point counts follow from the hops. An operator that reaches k hops on the fine lattice couples coarse sites whose coordinates differ by at most 1 in at most k directions. For $k = 1$ and 2 this gives $1 + 8 = 9$ and $1 + 8 + 24 = 33$ points, and for $k \geq 4$ all $3^4 = 81$.

The correction. By $S^{-1} = \pi D^{-1} \iota$, the inverse of the right form is

$$\tilde{A}_R^{-1} = C_e\, S^{-1}\, (S_{\rm PV}^\dagger)^{-1}\, C_{{\rm PV},e}^\dagger = C_e\, \pi D^{-1} \iota\; \pi (D_{\rm PV}^\dagger)^{-1} \iota\; C_{{\rm PV},e}^\dagger, \tag{3.1}$$

a product of two inverses of one-hop operators and site-local factors. The Galerkin operator of $\tilde{A}_R$ itself would have a large coarse stencil, since $\tilde{A}_R$ reaches four hops. The Galerkin operators of D and $D_{\rm PV}$ are nearest-neighbor, since D is nearest-neighbor in four dimensions and every aggregate contains the whole fifth dimension. We therefore replace each inverse by its Galerkin approximation $\mathcal{P}(\mathcal{P}^\dagger X \mathcal{P})^{-1}\mathcal{P}^\dagger$. For both forms and any α, the correction of a residual r on the even sites is

$$e = O_L\, \pi\mathcal{P}\, \mathcal{D}_{\rm c}^{-1} \mathcal{P}^\dagger \iota\;\; O_M\;\; \pi\mathcal{P}\, \left(\mathcal{P}^\dagger D_{\rm PV}^\dagger \mathcal{P}\right)^{-1} \mathcal{P}^\dagger \iota\;\; O_R\, r, \qquad \mathcal{D}_{\rm c} = \mathcal{P}^\dagger D \mathcal{P}, \tag{3.2}$$

with the site-local factors O_R, O_M, O_L of Table 2. They come from writing each $\tilde{A}^{-1}$ as in Eq. (3.1). For example, for the left form with α in both operators, Eq. (2.26) gives

$$\tilde{A}_L^{(\alpha)\,-1} = Q_e^{-1}\, S^{-1}\, C_e Q_e^2 C_{{\rm PV},e}^\dagger\, (S_{\rm PV}^\dagger)^{-1}\, Q_e^{-1}, \tag{3.3}$$

so $O_R = Q_e^{-1}$, $O_M = C_e Q_e^2 C_{{\rm PV},e}^\dagger$ and $O_L = Q_e^{-1}$. The two coarse inverses are those of the plain D and $D_{\rm PV}$; within the correction, α is in these factors only. Figure 3 follows a residual through Eq. (3.2). It moves between the even sites, the whole lattice and the coarse lattice twice, and the result of the first coarse solve, taken on the even sites, is the right-hand side of the second.

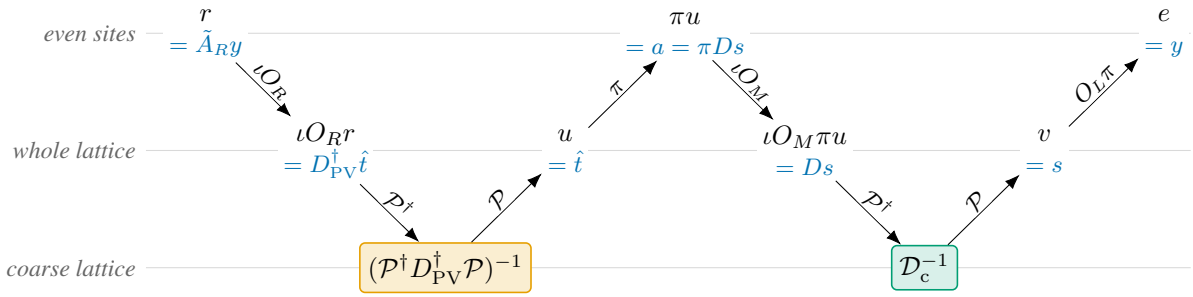


FIG. 3. One factorized coarse correction, from r (left) to e (right), which is Eq. (3.2) read from the right, on its three lattices (the Pauli–Villars coarse solve in orange, the quark one in green): $u = \mathcal{P}(\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P})^{-1}\mathcal{P}^\dagger \iota O_R r$ is the result of the first coarse solve, that of the adjoint Pauli–Villars operator, and its even part, with O_M, is the right-hand side of the second, that of the quark operator, whose result is $v = \mathcal{P}\mathcal{D}_{\mathrm{c}}^{-1}\mathcal{P}^\dagger \iota O_M \pi u$; $e = O_L \pi v$. In blue, after "=", the values for $r = \tilde{A}_R y$ in the right form with exact coarse solves and $s, \hat{t} \in \operatorname{range} \mathcal{P}$ (Eq. (3.8); $O_R = C_{\mathrm{PV},e}^\dagger$, $O_M = 1$, $O_L = C_e$): the first solve returns the lift $\hat{t}$, whose even part is $a = \pi D s$, the second returns the lift s, and the correction returns y.

TABLE 2. Site-local factors of the correction (3.2). They follow from $\tilde{A}^{-1}$ with $S^{-1} = \pi D^{-1} \iota$; with α the site-diagonal blocks are $C_e Q_e$ and $Q_e C_{\mathrm{PV},e}^\dagger$.

form	O_R	O_M	O_L
right, any α	$C_{\mathrm{PV},e}^\dagger$	1	C_e
left, α in D only	1	$C_e Q_e C_{\mathrm{PV},e}^\dagger$	Q_e^{-1}
left, α in D and D_{PV}	Q_e^{-1}	$C_e Q_e^2 C_{\mathrm{PV},e}^\dagger$	Q_e^{-1}

2. *Aggregation and prolongation*

We divide the fine lattice into rectangular four-dimensional blocks. Each block contains the whole fifth dimension and becomes one coarse site. We split each test vector v into its Γ_5 eigencomponents $v_\pm = \frac{1}{2}(1 \pm \Gamma_5)v$ and orthonormalize the $+$ parts and the $-$ parts separately within each block. With n_v test vectors a coarse site then carries $2n_v$ components: the first n_v of positive and the last n_v of negative Γ_5-chirality.

This split makes $\mathcal{P}$ compatible with Γ_5. Each column of $\mathcal{P}$ is an eigenvector of Γ_5 with eigenvalue $+1$ or -1, so with $\Sigma = \operatorname{diag}(\mathbf{1}_{n_v}, -\mathbf{1}_{n_v})$ on each coarse site,

$$\Gamma_5 \mathcal{P} = \mathcal{P}\Sigma. \tag{3.4}$$

It follows that $\mathcal{P}^\dagger(\Gamma_5 X \Gamma_5)\mathcal{P} = \Sigma\, \mathcal{P}^\dagger X \mathcal{P}\, \Sigma$ for any fine operator X. In particular $\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P} = \Sigma \mathcal{D}_{\mathrm{c}}^{\mathrm{PV}} \Sigma$ with $\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}} = \mathcal{P}^\dagger D_{\mathrm{PV}} \mathcal{P}$ whenever $\Gamma_5 D_{\mathrm{PV}} \Gamma_5 = D_{\mathrm{PV}}^\dagger$, that is for $c = 0$.

The split also halves the storage, for any c, since it uses only Γ_5 and no property of D. Since $\Gamma_5 = \gamma_5 R$, the condition $\Gamma_5 v_\pm = \pm v_\pm$ reads $v_\pm(L_s - 1 - s) = \pm \gamma_5 v_\pm(s)$. So each part is fixed by its values for $s < L_s/2$, and $\mathcal{P}$ is stored only there. On the even sites the transfer operators are $\mathcal{P}^\dagger \iota$ and $\pi \mathcal{P}$.

3. *The basis* $\{s, \hat{t}\}$

The prolongation is built from vectors y_j on the even sites that approximate low modes of $\tilde{A}_R$ (Sec. III.C). Two facts shape how we use them.

- $\mathcal{P}$ and the coarse operators act on the whole lattice, so each y must be extended, or *lifted*, to the whole lattice. We fill the odd sites so that the fine operator maps the lifted vector to the even-odd quantity, just as ψ_o is rebuilt from ψ_e after an even-odd solve.

- The correction (3.2) contains two Galerkin solves, one with $D_{\mathrm{PV}}^\dagger$ and one with D. Each is exact only on vectors in $\operatorname{range}\mathcal{P}$. So we lift each y in two ways, one for each solve.

The two lifts are

$$s = \big(C_e^{-1}y,\ -C_o^{-1}D_{oe}C_e^{-1}y\big), \tag{3.5}$$

$$\hat{t} = \big(a,\ -(C_{\mathrm{PV},o}^\dagger)^{-1}(D_{\mathrm{PV},eo})^\dagger a\big), \qquad a = \pi D s. \tag{3.6}$$

The lift s is chosen so that Ds vanishes on the odd sites. With the blocks of Eq. (2.22),

$$Ds = \begin{pmatrix} C_eC_e^{-1}y - D_{eo}C_o^{-1}D_{oe}C_e^{-1}y \\ D_{oe}C_e^{-1}y - C_oC_o^{-1}D_{oe}C_e^{-1}y \end{pmatrix} = \begin{pmatrix} SC_e^{-1}y \\ 0 \end{pmatrix} = \iota\, SC_e^{-1}y. \tag{3.7}$$

In the same way $\hat{t}$ is chosen so that $D_{\mathrm{PV}}^\dagger\hat{t}$ vanishes on the odd sites. The block form of $D_{\mathrm{PV}}^\dagger$ has $(D_{\mathrm{PV},eo})^\dagger$ as its odd-even block and $C_{\mathrm{PV},o}^\dagger$ as its odd diagonal block, and the same computation gives $D_{\mathrm{PV}}^\dagger\hat{t} = \iota\, S_{\mathrm{PV}}^\dagger a$.

The first coarse solve of the correction returns $\hat{t}$ and the second returns s (Eq. (3.8) below). Since $\hat{t}$ is made from Ds, the two lifts follow the order of the solves. With the y alone in the basis, each factor of the correction would be an approximation, and the errors of the two would multiply. The $2n$ vectors $\{s_j, \hat{t}_j\}$ are the test vectors, $n_v = 2n$.

The two lifts differ in the fifth dimension. Their odd parts are built with different inverses, C_o^{-1} for s and $(C_{\mathrm{PV},o}^\dagger)^{-1}$ for $\hat{t}$. By Eq. (2.17), C^{-1} carries $P_+\psi$ from slice s to $s+1$ with the factor $q = d_-/d_+$ and $P_-\psi$ the other way, while $C^\dagger$ is its transpose and carries both in the opposite direction; the hops D_{oe} and $(D_{\mathrm{PV},eo})^\dagger$ likewise contain Y and $Y^\dagger$. In each chirality the odd components of s and $\hat{t}$ are therefore weighted toward opposite ends of the fifth dimension, and a space built from the s_j alone does not contain the $\hat{t}_j$ that the first coarse solve, with $D_{\mathrm{PV}}^\dagger$, returns.

The structure is familiar from block preconditioners for the incompressible Navier–Stokes equations [53]. There the velocity is eliminated, as the odd sites are here, and the pressure Schur complement is approximated by a product of simpler operators, each solved by its own multigrid. In the least-squares commutator preconditioner, $(BB^{\mathsf{T}})^{-1}(BFB^{\mathsf{T}})(BB^{\mathsf{T}})^{-1}$, these are two Poisson-like inverses. The product is only as good as each factor on the vectors it is applied to; here those are $\hat{t}$ and s.

Exactness for the right form. Let $s, \hat{t} \in \operatorname{range}\mathcal{P}$, let $\mathcal{D}_{\mathrm{c}}$ and $\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger\mathcal{P}$ be invertible, and let both coarse systems in Eq. (3.2) be solved exactly, with the exact adjoint $\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger\mathcal{P} = (\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}})^\dagger$. Then the right-form correction returns the vector it is built from,

$$e(\tilde{A}_R y) = y. \tag{3.8}$$

For any invertible $\mathcal{P}^\dagger X\mathcal{P}$ the map $\mathcal{P}(\mathcal{P}^\dagger X\mathcal{P})^{-1}\mathcal{P}^\dagger X$ is the identity on $\operatorname{range}\mathcal{P}$, and we follow $r = \tilde{A}_R y$ through the correction with the factors of the right form, $O_R = C_{\mathrm{PV},e}^\dagger$, $O_M = 1$, $O_L = C_e$. First, $O_R r = C_{\mathrm{PV},e}^\dagger\tilde{A}_R y = S_{\mathrm{PV}}^\dagger SC_e^{-1}y = S_{\mathrm{PV}}^\dagger a$ and $\iota S_{\mathrm{PV}}^\dagger a = D_{\mathrm{PV}}^\dagger\hat{t}$, so the first coarse solve returns $\hat{t}$, whose even part is $\pi\hat{t} = a = \pi Ds$. The second solve then has the right-hand side $\mathcal{P}^\dagger\iota\pi Ds = \mathcal{P}^\dagger Ds$, because Ds vanishes on the odd sites, and returns s, whose even part is $\pi s = C_e^{-1}y$. Finally $O_L = C_e$ gives $e = y$.

In practice. Equation (3.8) needs three conditions that hold only approximately. The stored $\mathcal{P}$ contains s and $\hat{t}$ only up to the rounding of its reduced-precision storage. The Σ-adjoint $\Sigma\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}\Sigma$ of Sec. III.B equals the exact adjoint only if $\Gamma_5 D_{\mathrm{PV}}\Gamma_5 = D_{\mathrm{PV}}^\dagger$, which fails for $c \neq 0$ by Eq. (2.10). And we stop the coarse solves at a loose tolerance. With both coarse factors solved on their exact adjoints to $|r|^2/|b|^2 < 10^{-24}$ in double precision and $\mathcal{P}$ in half precision, the correction of $r = \tilde{A}_R y_j$ returns y_j to $\|e - y_j\|/\|y_j\| = 3\times10^{-4}$–$5\times10^{-4}$ on $4^3\times8$ (2^4 aggregates, $M_5 = 1.8$, $m = 0.05$) and 3×10^{-4}–2.5×10^{-3} on 12^4 (setup S, $m = 0.001$), against 3–37 for random vectors. So Eq. (3.8) holds to the

rounding of $\mathcal{P}$, about 10^{-3}. The Σ-adjoint costs no measurable iterations (Sec. V.C.3), and accurate coarse solves are not faster with GCR(2) and stagnate on 12^4 (Secs. V.D.4, V.C.4).

The left form. For the left form with α in both operators, the same argument needs the quark lift s' of $Q_e y'$ and the Pauli–Villars lift $\hat{t}'$ of $u = C_{\mathrm{PV},e}^{\dagger\,-1} Q_e^{-2} C_e^{-1} S Q_e y'$ in range $\mathcal{P}$ (Table 2 gives $O_M u = S Q_e y'$). For $\alpha = 1$ and $y' = C_e^{-1} y$ we have $s' = s$. But $u = C_{\mathrm{PV},e}^{\dagger\,-1} C_e^{-1} a$ differs from a by a site-local matrix that mixes s and spin, so $\hat{t}'$ is in general not in the span of the right-form vectors, and the correction is not exact for the left form. For $\alpha = 1$ and $r = \tilde{A}_L y'$ with $y' = C_e^{-1} y_j$ the correction misses y' by $\|e - y'\|/\|y'\| = 0.05$–0.18 on $4^3 \times 8$ and 1.6–24 on 12^4, not much better than for random vectors (8–35). The left form converges nonetheless: it has the spectrum of the right form and needs one to five iterations more or fewer (Secs. V.D.2, V.D.4, V.C.4). So the cycle does not need the exactness (3.8), but with the same vectors y it does need $\hat{t}$ in the basis; on $24^3 \times 48$ one lift of twice as many vectors y does as well (Sec. V.C.2).

4. The coarse operators

We assemble $\mathcal{D}_\mathrm{c} = \mathcal{P}^\dagger D \mathcal{P}$ and $\mathcal{D}_\mathrm{c}^\mathrm{PV} = \mathcal{P}^\dagger D_\mathrm{PV} \mathcal{P}$ from the hops of the fine operator between neighboring blocks and from its part within a block. Each coarse site has a $2n_v \times 2n_v$ diagonal block and one link block per direction, nine blocks in all. We do not assemble the adjoint factor $\mathcal{P}^\dagger D_\mathrm{PV}^\dagger \mathcal{P}$. We apply it either exactly, as the conjugate transpose $(\mathcal{D}_\mathrm{c}^\mathrm{PV})^\dagger$ of the stored blocks, or approximately as $\Sigma \mathcal{D}_\mathrm{c}^\mathrm{PV} \Sigma$. When all extents of the coarse lattice are even, the coarse hops connect coarse sites of opposite parity, and both coarse operators have an even-odd form of their own.

B. The coarse solvers

We solve both coarse systems only approximately, in reduced precision, to a loose relative residual or a maximum number of iterations. Reaching the tolerance is not required. The two are solved in sequence: first the Pauli–Villars factor, whose solution enters the right-hand side of the quark factor. The default, used for all results unless stated, is BiCGStab on the even-odd form of the quark factor and MINRES on the Σ-Hermitian form of the Pauli–Villars factor, both to $|r|^2/|b|^2 < 10^{-2}$ (Algorithms 8 and 5, Appendix D). This subsection describes these and the alternatives.

1. The quark factor

Write $\mathcal{D}_\mathrm{c} = \begin{pmatrix} A_{ee} & B_{eo} \\ B_{oe} & A_{oo} \end{pmatrix}$ on the coarse lattice. The system $\mathcal{D}_\mathrm{c} c = g$ is solved through its Schur complement,

$$\left(A_{ee} - B_{eo} A_{oo}^{-1} B_{oe}\right) c_e = g_e - B_{eo} A_{oo}^{-1} g_o, \qquad c_o = A_{oo}^{-1} (g_o - B_{oe} c_e), \tag{3.9}$$

with BiCGStab or truncated flexible GCR, preconditioned by the exact inverses of the diagonal blocks A_{ee}^{-1}. We need a non-Hermitian Krylov method here: $\mathcal{D}_\mathrm{c}$ is not Hermitian, and $\Sigma \mathcal{D}_\mathrm{c}$ is Hermitian only for $c = 0$. BiCGStab [21] is Algorithm 8 of Appendix D.

2. The Pauli–Villars factor

The system is $(\mathcal{P}^\dagger D_\mathrm{PV}^\dagger \mathcal{P})\, w = g$. Two treatments are available.

Exact adjoint. Solve $(\mathcal{D}_\mathrm{c}^\mathrm{PV})^\dagger w = g$ with a non-Hermitian Krylov method, preconditioned by the inverses of the adjoint diagonal blocks, $(A_{xx}^\dagger)^{-1} = (A_{xx}^{-1})^\dagger$.

Σ-Hermitian form. Replace the adjoint by $\Sigma \mathcal{D}_\mathrm{c}^\mathrm{PV} \Sigma$ and solve the equivalent system

$$\mathcal{H}_\mathrm{c}\, w = \Sigma g, \qquad \mathcal{H}_\mathrm{c} = \mathcal{D}_\mathrm{c}^\mathrm{PV} \Sigma, \tag{3.10}$$

with MINRES. $\mathcal{H}_\mathrm{c}$ is Hermitian (indefinite) exactly when $(\mathcal{D}_\mathrm{c}^\mathrm{PV})^\dagger = \Sigma \mathcal{D}_\mathrm{c}^\mathrm{PV} \Sigma$, that is for $c = 0$ by Eqs. (2.10) and (3.4). For $c \neq 0$ MINRES works on a non-Hermitian matrix and has no guarantee. On the

coarse space the non-Hermitian part is a few percent, up to 14% in the directions of the smallest singular values, and linear in c. Even so, MINRES on $\mathcal{H}_c$ needs the same outer iterations at $c = 0.5$ as for the Shamir operator and is better than the exact adjoint with BiCGStab (Sec. V.C.3). The absolute-value block preconditioner and the additive level preconditioner below, also exact only for $c = 0$, are used as they are.The preconditioner must be Hermitian positive definite; we use the absolute value of the diagonal blocks [48], $|\mathcal{H}_{xx}|^{-1} = V|\Lambda|^{-1}V^\dagger$ with $\frac{1}{2}(\mathcal{H}_{xx} + \mathcal{H}_{xx}^\dagger) = V\Lambda V^\dagger$ on each coarse site. On a coarse lattice with even extents, the Schur complement of $\mathcal{H}_c$ on the even sites is $S_c\Sigma_e$, with S_c the Schur complement of $\mathcal{D}_c^{\mathrm{PV}}$; one solves for $u_e = \Sigma w_e$ on the even sites, completes u on the odd sites as in Eq. (3.9), and sets $w = \Sigma u$.

3. *Levels below the coarse lattice*

Both factors may be preconditioned by further levels. The coarse sites are grouped into super-blocks; on each super-block, local vectors are formed from the coarse test vectors $\mathcal{P}^\dagger s_j$ and $\mathcal{P}^\dagger \hat{t}_j$, split by Σ and orthonormalized (Gram–Schmidt applied twice); the level operators are obtained by probing the nearest-neighbor stencil; the last level is inverted densely. With Q_ℓ the prolongation into level $\ell - 1$ and B_ℓ the diagonal blocks of level ℓ:

Quark factor. A V-cycle with the coarse correction followed by one block-Jacobi sweep on the updated residual, without pre-smoothing,

$$z = Q_{\ell+1}M_{\ell+1}Q_{\ell+1}^\dagger r, \qquad M_\ell r = z + B_\ell^{-1}(r - \mathcal{D}_\ell z); \tag{3.11}$$

on the coarse lattice itself the same multiplicative combination is used, and with the even-odd coarse form its even part preconditions the Schur system. The level test vectors can optionally be refined by a few GCR steps preconditioned by the levels already built.

Pauli–Villars factor (Σ-Hermitian form only). The additive Hermitian positive preconditioner

$$M_\ell = |B_\ell|^{-1} + Q_{\ell+1}M_{\ell+1}Q_{\ell+1}^\dagger, \qquad M_{\ell_{\max}} = |\mathcal{H}_{\ell_{\max}}|^{-1}, \tag{3.12}$$

in the spirit of absolute-value preconditioning [48].

Levels instead of deflation. The levels below compress the low modes further by aggregation and invert the last level exactly, by LU in double precision (cuSOLVER on the device), so every mode that reaches that level is treated without computing any eigenvector. The alternative is to deflate the coarse factors with a few of their eigenvectors. With the basis $\{s, Ds\}$ in place of $\{s, \hat{t}\}$ (12^4, quenched, $\beta = 6.0$, $m = 0.005$), the 32 Ritz pairs closest to zero from a Lanczos run of 600 steps with full reorthogonalization had relative residuals up to 2; deflating with them reduced the MINRES steps of the Pauli–Villars factor from 402 to 350, and the solution stalled at a residual of 9×10^{-3}. With $\{s, \hat{t}\}$, whose coarse operators differ (Sec. V.C.2), coarse deflation remains to be tried. Boyle and Oare [43] report that deflating their coarse operator did not work, and add a large coarsest level that is inverted densely.

C. Setup

The setup produces the vectors $y_1, \ldots, y_n$ on the even sites, then the basis $\{s_j, \hat{t}_j\}$, $\mathcal{P}$ and the coarse operators. It always uses the right form; the same setup serves the left forms and every α (Sec. III.A). It has two stages. A Chebyshev filter finds approximate low modes without any multigrid. A bootstrap then improves them with the cycle built from them, so that they also cover what the first coarse space misses.

1. *Chebyshev-filtered subspace iteration*

The vectors approximate the right singular vectors of the smallest singular values of $\tilde{A}_R$, that is the low eigenvectors of the Hermitian positive operator $H_n = \tilde{A}_R^\dagger \tilde{A}_R$. We find them by filtering. Let T_d be the Chebyshev polynomial of degree d and $\ell(x) = \big(2x - (\lambda_{\mathrm{hi}} + \lambda_{\mathrm{lo}})\big)/(\lambda_{\mathrm{hi}} - \lambda_{\mathrm{lo}})$. Then $|T_d(\ell(x))| \le 1$ on $[\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}]$, while it grows quickly below λ_{lo}. So $T_d(\ell(H_n))$ amplifies the low modes against the rest. We apply it by the three-term recurrence.

Algorithm 1. LOWMODES: Chebyshev-filtered subspace iteration on $H_n = \tilde{A}_R^\dagger \tilde{A}_R$.

$X \leftarrow n$ random vectors on the even sites
$\lambda_{\rm hi} \leftarrow$ a bound slightly above $\lambda_{\max}(H_n)$ (power iteration); $\lambda_{\rm lo} \leftarrow$ a small fraction of $\lambda_{\rm hi}$
for each round **do**
$d \leftarrow$ a degree proportional to $\sqrt{\lambda_{\rm hi}/\lambda_{\rm lo}}$, within fixed bounds
$X \leftarrow T_d(\ell(H_n))\, X$; orthonormalize X through its Gram matrix
$G \leftarrow X^\dagger H_n X = V \Theta V^\dagger$; $X \leftarrow XV$ ▷ Rayleigh–Ritz
$\lambda_{\rm lo} \leftarrow \theta_{\max}$ ▷ largest Ritz value
end for
return X, Θ

2. *Bootstrap*

Each bootstrap round has three steps. It builds the basis and the two-grid cycle of Sec. III.D from the current y. It replaces every y_j by a few FGMRES steps (Algorithm 6) on $\tilde{A}_R x = y_j$ from $x = 0$, preconditioned by that cycle. And it orthonormalizes the new y through their Gram matrix ($y \leftarrow yV\Lambda^{-1/2}$ with $y^\dagger y = V\Lambda V^\dagger$). The cycle damps the components it already handles, so the new vectors are enriched in what it does not.

Algorithm 2. Setup of the two-grid method.

$y \leftarrow$ LOWMODES(H_n)
for each bootstrap round **do**
build $\mathcal{P}$ from $\{s_j, \hat{t}_j\}$ of y and the cycle $\mathcal{M}$ on it
for $j = 1, \ldots, n$ **do**
$y_j \leftarrow$ a few FGMRES steps on $\tilde{A}_R x = y_j$, preconditioned by $\mathcal{M}$
end for
orthonormalize y
end for
compute $s_j, \hat{t}_j$ by Eqs. (3.5)–(3.6); split by Γ_5; orthonormalize per block $\to \mathcal{P}$
assemble $\mathcal{D}_{\rm c}$, $\mathcal{D}_{\rm c}^{\rm PV}$, the inverses of their diagonal blocks, their even-odd forms
optionally, build the levels below from $\mathcal{P}^\dagger s_j$, $\mathcal{P}^\dagger \hat{t}_j$

D. The Chebyshev smoother and the cycle

1. *Smoother*

The smoother is ν steps of Chebyshev iteration [49, 50] for $\tilde{A}x = b$ from $x = 0$, i.e. a fixed polynomial $x = p(\tilde{A})b$ of degree $\nu - 1$. It is defined by an ellipse with real foci $[\lambda_{\rm lo}, \lambda_{\rm hi}]$: $\lambda_{\rm hi}$ slightly above an estimate of the spectral radius of $\tilde{A}$ from a power iteration, and $\lambda_{\rm lo} = \lambda_{\rm hi}/\kappa_s$ for a chosen ratio κ_s. With $\theta = \frac{1}{2}(\lambda_{\rm hi} + \lambda_{\rm lo})$, $\delta = \frac{1}{2}(\lambda_{\rm hi} - \lambda_{\rm lo})$, $\sigma = \theta/\delta$, $\varrho_0 = 1/\sigma$, $r_0 = b$ and $d_0 = r_0/\theta$, step k is

$$x \leftarrow x + d_k, \quad r_{k+1} = r_k - \tilde{A} d_k, \quad \varrho_{k+1} = \frac{1}{2\sigma - \varrho_k}, \quad d_{k+1} = \varrho_{k+1}\varrho_k\, d_k + \frac{2\varrho_{k+1}}{\delta}\, r_{k+1}, \tag{3.13}$$

The last step only updates x, so ν steps cost $\nu - 1$ applications of $\tilde{A}$. The recurrence keeps four fine vectors, x, r, d and $\tilde{A}d$, whatever ν, and needs no inner products; the smoother is the same linear map at every application. A Krylov smoother, such as the fixed number of GCR steps of the Pauli–Villars multigrid of Ref. [40], keeps two more fine vectors per step, just as the outer flexible GCR does, and in a fixed amount of memory it lowers the largest volume that can be solved (on $24^3 \times 48$ the multigrid with outer GCR(2) already peaks at 20.2 of 24 GB of device memory, and outer GCR(8) does not fit, Sec. V.D.4). We use only a real focal interval. The ratio κ_s is not derived from the operator; it has to be chosen. The polynomial damps the part of the spectrum above λ_{lo} and leaves the rest to the coarse correction, and an interval that is too wide or too narrow can make the outer solve stagnate. The interval that works differs between the two setups of Sec. V (Secs. V.A and V.C.4).

2. Cycle

Algorithm 3. Two-grid cycle $y = \mathcal{M}b$ for $\tilde{A}$.

1: $y \leftarrow p(\tilde{A})\, b$ ▷ pre-smoothing
2: $r \leftarrow b - \tilde{A}y$
3: $y \leftarrow y + e(r)$ ▷ correction (3.2)
4: $r \leftarrow b - \tilde{A}y$
5: $y \leftarrow y + p(\tilde{A})\, r$ ▷ post-smoothing

One cycle costs 2ν applications of $\tilde{A}$ (two smoothings and two residuals) plus the two coarse solves. Because the coarse solves are Krylov iterations stopped by a residual test, $\mathcal{M}$ is not a fixed linear map.

E. The outer solver

The outer iteration solves $\tilde{A}y = \rho$ with the cycle as a right preconditioner, and we rebuild ψ as in Sec. II.C. The cycle $\mathcal{M}$ changes from one application to the next, so the outer method must be flexible: FGMRES or flexible GCR (Algorithms 6 and 4, Appendix D), which keep the preconditioned directions. BiCGStab with the cycle is also available, but its convergence theory has no flexible counterpart.

Flexibility alone gives no convergence guarantee. For a fixed $\mathcal{M}$, restarted or truncated methods converge for every restart length if the field of values of $\tilde{A}\mathcal{M}$ lies in the right half-plane [44]. Otherwise only the full method is covered, by the spectrum and the pseudospectra of $\tilde{A}\mathcal{M}$ [35, 51], and a truncated method can stagnate [52]. Secs. V.D.1–V.D.4 examine this for the present cycle. We monitor convergence on the even-odd system and evaluate the residual of the original equation, $\|D\psi - \eta\|/\|\eta\|$, with the plain D after reconstruction. The reference solver is CG on the normal equation of the even-odd operator $C_e^{-1}S$ (CGNE). It is robust but squares the condition number. It runs in double precision, or in mixed precision: CG in single precision with reliable updates in double precision. Appendix D compares it with CG on $S^\dagger S$. The solves with $D^\dagger$ that HMC needs use the same cycle between two Γ_5 (Sec. IV).

IV. Solving with $D^\dagger$

HMC needs $(D^\dagger D)^{-1}$ at every force evaluation, so it solves with $D^\dagger$ as often as with D. The multigrid of D also solves with $D^\dagger$, with no second setup, coarse operator or cycle. The construction needs only an approximate inverse of D, so it applies to any preconditioner built for the unsquared operator. The tool is the exact relation (2.13) between D and $D^\dagger$.

A. The idea

For Wilson fermions $D_W^\dagger = \gamma_5 D_W \gamma_5$, so $D_W^{-\dagger} = \gamma_5 D_W^{-1} \gamma_5$, and a solver for D_W solves with $D_W^\dagger$ as well. For Möbius fermions this fails at non-zero c (Sec. II.B.2). We use $\Gamma_5 D^{-1} \Gamma_5$ anyway, as a preconditioner for $D^\dagger$. The question is how good a preconditioner it is.

Multiply $D^\dagger$ by $\Gamma_5 D^{-1}\Gamma_5$ and use $D^\dagger = \Gamma_5 D_-^{-1} D\, \Gamma_5 D_-^\dagger$ from Eq. (2.13) and $\Gamma_5 D_-^\dagger \Gamma_5 = D_-$:

$$D^\dagger\, \Gamma_5 D^{-1}\Gamma_5 = \Gamma_5 D_-^{-1} D\, D_- D^{-1}\, \Gamma_5 = \Gamma_5 \big(1 + D_-^{-1}[D, D_-] D^{-1}\big)\Gamma_5. \tag{4.1}$$

The commutator follows from $D = D_+ - D_- Y$ and $[D_+, D_-] = 0$: $[D, D_-] = -D_-[Y, D_-] = -c\, D_-[Y, W]$. The factor D_- cancels, and

$$\boxed{D^\dagger\, \Gamma_5 D^{-1}\Gamma_5 = \Gamma_5\, (1 - N)\, \Gamma_5, \qquad N = c\, [Y, W]\, D^{-1}.} \tag{4.2}$$

Equation (4.2) has three properties used below:

- D_-^{-1} has dropped out. We never invert D_-, which we cannot do (Sec. IV.D).
- $[Y, W] = -\frac{1}{2}[Y, K]$ is a bounded operator: one hop in four dimensions and one in the fifth. So N is large only where D^{-1} is large, on the low modes of D.
- On a low mode v the commutator acts through the γ_μ in K, which flip chirality. A low mode of D carries on each wall the chirality bound there, P_- at $s = 0$ and P_+ at $s = L_s - 1$; $[Y, W]v$ carries on each wall the opposite chirality, which no low mode has there. So N takes the low modes out of their span, and N^2 is small. Then $(1 - N)^{-1} \approx 1 + N$, and a Krylov method needs only a few steps more than for the identity.

The third remark is a heuristic; Sec. V.B.2 tests it and finds that it fails on the smallest smeared lattice (12^4 in setup S).

B. On the even sites

The solves of HMC are on the even sites, so we need the same idea for the Schur complement S. $D^\dagger$ has the Schur complement $S^\dagger$ (Sec. IV.C), and $S^{-1} = \pi D^{-1} \iota$. Because Γ_5 acts within a site, it commutes with π and ι, and

$$S^{-\dagger} = \pi D^{-\dagger} \iota \approx \pi\, \Gamma_5 D^{-1}\Gamma_5\, \iota = \Gamma_5 S^{-1}\Gamma_5. \tag{4.3}$$

The solve of $D^\dagger w = r$ then goes as follows.

1. Form $r'_e = r_e - D_{oe}^\dagger (C_o^\dagger)^{-1} r_o$, so that $S^\dagger w_e = r'_e$.
2. Write $w_e = (C_e^\dagger)^{-1} z$. Then $\mathcal{K}^\dagger z = r'_e$ with $\mathcal{K} = C_e^{-1} S$, the operator HMC already uses.
3. Solve for z by flexible GCR with the preconditioner $C_e^\dagger\, \Gamma_5 S^{-1}\Gamma_5$. The inverse S^{-1} is one application of the cycle $\mathcal{M}$ of $\tilde{A}_R$: $S^{-1} t = C_e^{-1} \tilde{A}_R^{-1} (S_{\rm PV} C_{{\rm PV},e}^{-1})^\dagger t \approx C_e^{-1}\, \mathcal{M}\, (S_{\rm PV} C_{{\rm PV},e}^{-1})^\dagger t$.
4. Recover $w_e = (C_e^\dagger)^{-1} z$ and $w_o = (C_o^\dagger)^{-1}(r_o - D_{eo}^\dagger w_e)$.

Each GCR step costs one cycle, as a step of the solve with D does.

C. The system on the even sites

For completeness we derive the even-site system of $D^\dagger$. The block form of $D^\dagger$ is

$$D^\dagger = \begin{pmatrix} C_e^\dagger & D_{oe}^\dagger \\ D_{eo}^\dagger & C_o^\dagger \end{pmatrix}, \tag{4.4}$$

because the (e, o) block of $D^\dagger$ is the adjoint of the (o, e) block of D. Eliminating the odd sites of $D^\dagger w = r$ as in Sec. II.C gives $\big(C_e^\dagger - D_{oe}^\dagger (C_o^\dagger)^{-1} D_{eo}^\dagger\big) w_e = r'_e$, and the matrix is $S^\dagger$ term by term.

TABLE 3. The two setups and the multigrid of the mass scan. All runs: Möbius $(b, c) = (1.5, 0.5)$, $L_s = 8$, $m_{\rm PV} = 1$, $\alpha_{\rm PV} = 1$, boundary $(1, 1, 1, -1)$, $\alpha = 0.6$ unless stated. Configurations: quenched, plaquette action at $\beta = 6.0$, on 12^4, 16^4, 24^4 and $24^3 \times 48$, one each; their plaquettes, t_0 and topological charges are in Appendix C.

	setup S (stout)	setup B (bare)
M_5	1.0	1.8
gauge smearing	stout, $\rho = 0.1$, 3 steps	none
Chebyshev smoother ν, κ_s	8, 10	12, 30
multigrid (both setups)		
aggregates	4^4 ($24^3 \times 48$: levels below with super-blocks 2, 3)	
test vectors	32 (16 filtered low modes of $\tilde{A}_R^\dagger \tilde{A}_R$, lifted to $\{s, \hat{t}\}$)	
setup	filter 2 rounds; bootstrap 2 rounds of 4 FGMRES steps	
coarse solves	single precision, $\lvert r\rvert^2/\lvert b\rvert^2 < 10^{-2}$	
precision	$\mathcal{P}$ half, cycle single, outer solver double	
outer solver	GCR(2) on $\tilde{A}_L$ to $\lvert r\rvert^2/\lvert b\rvert^2 < 10^{-18}$	

D. Why D_- cannot be inverted

Eq. (2.13) seems to offer more: the exact null vectors of $D^\dagger$, and an exact reduction of $D^\dagger w = r$ to a solve with D. Both need D_-^{-1}, and D_- cannot be inverted in practice because of its spectrum. Since $D_- = cW - 1 = c\,(W - 1/c)$, D_- is c times the Wilson operator of mass $m_0 = -M_5 - 1/c$: -3 in setup S and -3.8 in setup B. The free eigenvalues of that operator,

$$m_0 + \sum_\mu (1 - \cos p_\mu) \pm i \Big(\sum_\mu \sin^2 p_\mu \Big)^{1/2}, \tag{4.5}$$

have real parts from m_0 to $m_0 + 8$, here from -3 to 5. Near real part zero they have imaginary parts of both signs, so the spectrum surrounds the origin. A Krylov method then has no low-degree polynomial p with $p(0) = 1$ that is small on the spectrum (Sec. II.A), and it makes no progress. Numerically (12^4, stout, $M_5 = 1$), GCR(8) on D_- and on $D_-^\dagger$ stops at $\lvert r\rvert^2/\lvert b\rvert^2 = 0.74$ (S) and 0.96 (B) after 20000 iterations, and BiCGStab diverges. The preconditioner of Eq. (4.2) avoids D_-^{-1} altogether.

V. Numerical results

A. Setups

Numerical setup. The gauge configurations are quenched, generated with the plaquette gauge action at $\beta = 6.0$, which corresponds to a lattice spacing of about 0.1 fm. The volumes are 12^4, 16^4, 24^4 and $24^3 \times 48$, with four configurations each (Table 23). Tables give means, medians or ranges over the four configurations as stated in their captions; the field of values, the pseudospectra, the comparison of the two even-odd forms (Appendix D) and the solvers without the multigrid use configuration 1. On them two setups of the Möbius operator are used (Table 3), both with $(b, c) = (1.5, 0.5)$, the choice of the JLQCD and the RBC/UKQCD Collaborations, and $L_s = 8$.

- *Setup B (bare links).* The domain-wall height $M_5 = 1.8$ is the value typical of practical simulations on unsmeared links, as in the RBC/UKQCD ensembles of Ref. [55]: on rough fields the lattice artifact of the Wilson kernel shifts the best value of M_5 above 1.

- *Setup S (stout links).* Three steps of the stout smearing of Sec. II.B.5 with $\rho = 0.1$ are applied to the links of the fermion operator, and $M_5 = 1.0$ is used, as in the JLQCD ensembles of Ref. [5] (its

supplemental material, which also gives $b + c = 2$, $b - c = 1$); the smearing largely suppresses the artifact that shifts the best value of M_5 away from 1.

The quark mass ranges over $m = 0.001$–0.1, and the Pauli–Villars mass is 1. Timings are from one NVIDIA data-center GPU with 24 GB of device memory; a second GPU was used for iteration counts and spectral quantities only. The device reductions add in an order that depends on the build and the GPU, so the iterations of otherwise equal runs can differ by a few. The setup seed is 21 in every run; Sec. V.D.4 shows how the count depends on it.

What the two setups mean for the multigrid. The residual breaking of chiral symmetry at finite L_s comes from the eigenvalues near zero of the Möbius kernel $H_M = \gamma_5(b + c)W\big(2 + (b - c)W\big)^{-1}$ [17], that is from near-zero modes of $W = D_W(-M_5)$, which on rough gauge fields come from small dislocations. The two setups keep them small in two ways, smoothing the field (S) or moving M_5 (B), and so give the multigrid low modes of different character. It shows in three measured places: the smoother needs the interval $[\lambda_{\mathrm{hi}}/10, \lambda_{\mathrm{hi}}]$ in S and $[\lambda_{\mathrm{hi}}/30, \lambda_{\mathrm{hi}}]$ in B, and with B's interval setup S stagnates, with S's setup B needs nearly three times the iterations (Sec. V.C.4); the field of values of $\tilde{A}_R$ reaches less far below zero with smearing ($\mathrm{Re}\, W \geq -4.6 \times 10^{-3}$ against -5.0×10^{-2} on 12^4, Sec. II.A); and the test vectors of $\tilde{A}_R$ are much closer to null vectors of $\tilde{A}_R^\dagger$ in S than in B ($\|\tilde{A}_R^\dagger y\|^2/\|y\|^2 \approx 0.015$ against 0.27 on 12^4 and 16^4, while $\|\tilde{A}_R y\|^2/\|y\|^2 \approx 0.003$ in both), which is why the dual cycle works in B but stalls in S (Sec. V.B.2, Table 8).

B. Performance

1. Comparison with CGNE

We compare the cost of the multigrid with that of the reference solver of Sec. III.E on $24^3 \times 48$, first for one random right-hand side over seven masses and then for whole point-source propagators. For one right-hand side and with the setup excluded, Ref. [40] found the Pauli–Villars multigrid an order of magnitude slower than red-black CG (Sec. I). Measured the same way, the present method is the faster solver at $m = 0.001$ from 24^4 on (Table 7).

Which even-odd CG. The reference solver is CG on $\mathcal{K}^\dagger\mathcal{K}$, $\mathcal{K} = C_e^{-1}S$, the even-odd operator of Bridge++, with Neff's α as for the multigrid. Boyle and Yamaguchi [40] used CG on $S^\dagger S$. For the Möbius setups of this work CG on $S^\dagger S$ needs 3–49% more iterations at $m = 0.001$ on $24^3 \times 48$ (Appendix D), and the multigrid is compared with the faster $\mathcal{K}^\dagger\mathcal{K}$.

Mixed-precision CGNE. For a fair comparison the reference solver also runs in mixed precision, CG in single precision with reliable updates in double precision; it needs 0.54–0.61 times the time of double precision at the same final residuals (Appendix D).

Mass dependence. The multigrid removes most of the critical slowing down of CGNE (Table 4, Fig. 4). On the four configurations, from $m = 0.1$ to $m = 0.001$ the iterations of the mixed-precision CGNE grow by a factor of 20–27 (S) and 18–21 (B), those of the multigrid by 1.6–4.1 (S) and 3.1–3.7 (B). At $m = 0.001$ the multigrid solve is 2.9–5.0 (S) and 4.8–6.0 (B) times faster than the mixed-precision CGNE, and its setup of about 210 s is paid back after 3.5–5.2 right-hand sides. The lowest factor, 2.9, is configuration 2 in setup S (Sec. V.D.4). The setup is paid back within one point-source propagator (12 columns) for $m \leq 0.005$. For $m \geq 0.05$ (S) the mixed-precision CGNE is faster or level, and at $m = 0.1$ (B) the two take nearly the same time. Against double-precision CGNE the solve at $m = 0.001$ is 3.9–9.4 (S) and 9.0–11.4 (B) times faster (Table 5). The multigrid needs three times the device memory of CGNE, which on a 24 GB GPU excludes GCR(k) with $k \geq 8$.

A point-source propagator. Whole propagators confirm the break-even estimates (Table 6). At $m = 0.001$ the multigrid takes 0.40–0.56 of the wall time of the mixed-precision CGNE, setup included, and a column is 3.0–6.2 times faster. At $m = 0.01$ the setup is not quite paid back within the 12 columns, as

TABLE 4. Mass scan on $24^3 \times 48$, means over the four configurations, one random right-hand side to $|r|^2/|b|^2 < 10^{-18}$ on the even-odd system: even-odd CGNE in double and in mixed precision against the multigrid (GCR(2) on $\tilde{A}_L$), $\alpha = 0.6$ for both and $\alpha = 1$ for the multigrid; iterations, solve time in seconds, coarse-operator applications, and N_{rhs}, the number of right-hand sides at which the multigrid setup (194–237 s) is paid back, $t_{\mathrm{setup}}/(t_{\mathrm{CGNE}} - t_{\mathrm{MG}})$, against the double- and the mixed-precision CGNE. The mixed-precision CGNE needs the iterations of the double-precision one within 1%, except at $m \leq 0.002$ (S), where it needs up to 30% more. Final $|D\psi - \eta|^2/|\eta|^2$: 9×10^{-18}–8×10^{-15} (CGNE, both), 2×10^{-19}–3×10^{-18} (MG). Device memory peak: CGNE 6.9 GB (double) and 7.2 GB (mixed), MG 20.2 GB ($\alpha = 0.6$). Table 5 gives the four configurations at $m = 0.001$.

	CGNE			MG, $\alpha = 0.6$			MG, $\alpha = 1$		N_{rhs}	
m	it	s	s, mixed	it	s	coarse	it	s	double	mixed
setup S ($M_5 = 1.0$, stout)										
0.1	182	5.0	2.9	14	6.1	1336	18	7.4	—	—
0.05	303	8.3	4.7	13	5.2	808	14	5.7	63	—
0.02	658	17.9	9.7	14	5.6	874	14	5.8	16	48
0.01	1174	31.9	17.1	16	6.7	1168	16	6.7	7.9	19
0.005	2079	56.5	30.1	20	8.3	1589	20	8.6	4.3	9.4
0.002	3424	92.9	50.4	26	11.7	2684	26	12.2	2.5	5.3
0.001	3668	99.5	61.3	32	15.4	4012	32	15.8	2.5	4.6
setup B ($M_5 = 1.8$)										
0.1	234	6.4	3.7	7	3.2	204	9	4.1	65	440
0.05	405	11.1	6.1	10	4.6	310	12	5.5	32	140
0.02	866	23.6	12.7	13	6.0	436	15	6.9	12	31
0.01	1481	40.2	21.5	16	7.2	575	17	7.9	6.3	15
0.005	2354	63.9	34.0	18	8.4	739	20	9.5	3.8	8.3
0.002	3628	98.5	52.3	22	10.3	1000	24	11.6	2.4	5.1
0.001	4456	120.9	64.2	24	11.5	1172	26	12.6	1.9	4.0

$N_{\mathrm{rhs}} \approx 15$–$19$ (Table 4) predicts. A point source needs more iterations than a random right-hand side, with either solver. The pseudoscalar correlators of the two agree to 4×10^{-8}–7×10^{-7} at $t \leq 3$ and to 2×10^{-6} at all 48 times. The multigrid stops on the residual of D ($|D\psi - \eta|^2/|\eta|^2 = 1$–$2 \times 10^{-18}$), the CGNE on that of its normal equation (6×10^{-17} to 3×10^{-16}). The device memory is 21.8 GB (multigrid) and 9.1 GB (mixed-precision CGNE).

Volume. Table 7 follows the solvers from 12^4 to $24^3 \times 48$ at $m = 0.001$ on quenched configurations of the same β. The multigrid iterations stay at 12–53, apart from one slow setup on 12^4, while those of the mixed-precision CGNE grow about 16 times in setup S and 8 times in B. On 12^4 the multigrid solve is slower than mixed CGNE (0.4–1.0 times its speed, the slow setup aside), on 16^4 it is 1.0–1.5 times faster in S and 1.6–2.1 times faster in B, and from 24^4 on it is 2.9–6.0 times faster. Its setup grows with the volume (6–8 s on 12^4, 204–218 s on $24^3 \times 48$), so the number of right-hand sides over which it is recovered depends on the volume as well. BiCGStab on $\tilde{A}_L$ alone (Sec. V.C.1) is slower than mixed CGNE on every volume.

Neff's α in the two setups. At $m = 0.001$, $\alpha = 0.6$ against $\alpha = 1$ saves CG on $\mathcal{K}^\dagger\mathcal{K}$ 35% (S) and 25% (B) of its iterations (Table 24) and the multigrid at most two (Table 4), as Sec. II.C expects. Over the masses the multigrid saves one to two iterations at every mass in setup B (from 9 to 7 at $m = 0.1$, from 26 to 24 at $m = 0.001$, means over the four configurations), and in setup S four at $m = 0.1$, about one at

TABLE 5. The four $24^3 \times 48$ configurations at $m = 0.001$, $\alpha = 0.6$ (Table 4): iterations and solve seconds, and the solve time of each CGNE over that of the multigrid.

		CGNE, double		CGNE, mixed		MG		speed-up	
setup	config.	it	s	it	s	it	s	double	mixed
S	1	3602	98.5	4090	59.9	28	13.2	7.5	4.5
S	2	3785	102.5	4936	76.1	53	26.0	3.9	2.9
S	3	3584	96.9	3622	51.9	23	10.3	9.4	5.0
S	4	3702	100.1	3990	57.2	25	12.3	8.1	4.7
B	1	4228	115.6	4232	62.0	26	12.9	9.0	4.8
B	2	4955	134.1	4959	70.9	25	11.8	11.4	6.0
B	3	4269	115.5	4276	61.1	22	10.3	11.2	5.9
B	4	4371	118.3	4377	62.7	23	10.9	10.9	5.8

TABLE 6. Point-source propagators (12 columns) on $24^3 \times 48$: the multigrid (setup seconds, iterations and seconds per column) against the mixed-precision CGNE (iterations and seconds per column), the wall time of the whole propagator, and the largest relative difference of the pseudoscalar correlator at $t = 0 \ldots 3$. At $m = 0.001$: ranges over the four configurations and the 12 columns; at $m = 0.01$: configuration 1.

		MG			CGNE, mixed		wall (s)		
	m	setup	it	s	it	s	MG	CGNE	$\lvert\Delta C\rvert/C$
S	0.001	201–216	26–59	11.7–29.4	3634–6400	52.1–91.6	381–562	678–1048	$\leq 1 \times 10^{-6}$
B	0.001	213–215	23–30	10.7–14.8	4354–5574	62.4–79.8	372–409	803–973	$\leq 7 \times 10^{-7}$
S	0.01	203	16–17	6.4–7.0	1337–1388	19.8–20.5	305	264	2×10^{-7}
B	0.01	215	17	8.1	1641–1745	24.2–25.7	331	324	4×10^{-8}

$m = 0.05$ and none from $m = 0.02$ down (32 for both at $m = 0.001$). The smoother intervals of Table 3 locate the difference. In B the smoother covers $[\lambda_{\text{hi}}/30, \lambda_{\text{hi}}]$, a wide band left to it at every mass, which α shortens. In S it covers $[\lambda_{\text{hi}}/10, \lambda_{\text{hi}}]$ and the coarse correction takes the rest; at light masses the low modes set the iterations, they are in the coarse space, and α has little left to improve. The spectra of the two setups would test this interpretation.

2. *Solves with* $D^\dagger$

Single solves. Table 8 compares three ways to solve with $D^\dagger$ from one setup of D. The right-hand sides are a random vector and the residual $r = b - D^\dagger D x_0$ after one sweep of the normal equations, solved to 10^{-10}. The second kind is the one that matters in HMC: its error lies mostly on the lowest modes.

- *The cycle of* $\tilde{A}_R^\dagger$ *on the same* $\mathcal{P}$ (the "dual cycle"): the factorized correction (3.2) with D and D_{PV} exchanged.
- *Flexible GCR on* $D^\dagger$ *with* $\Gamma_5 \mathcal{M} \Gamma_5$, Eq. (4.2), one cycle a step.

The preconditioner $\Gamma_5 \mathcal{M} \Gamma_5$ needs two to three iterations more than the solve with D, in both setups and on both kinds of right-hand side, as the heuristic predicts. More inner cycles per step do not reduce the count (9 iterations with one, two or four cycles on 16^4): the extra steps come from N, since a more exact $\mathcal{M}$ does not remove them. The dual cycle, in contrast, stalls on the hard right-hand side in setup S.

On 12^4 in setup S, $m = 0.001$, the residual right-hand side stalls GCR(2) on all four configurations, and GCR(8) needs 13–222 iterations. On 16^4, where configurations 2–4 have $Q = +1$, $+1$ and $+3$,

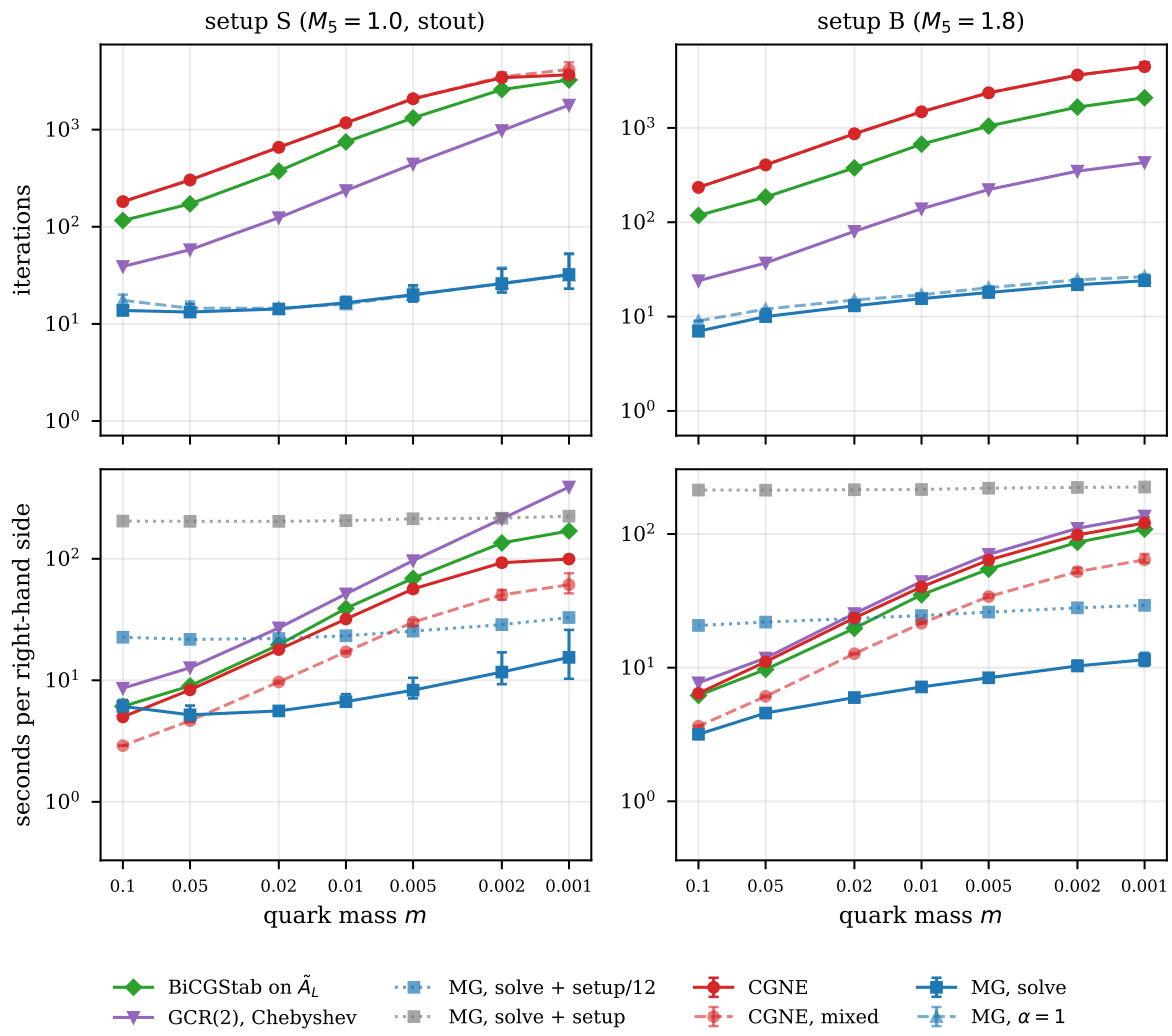


FIG. 4. The mass scan of Table 4, the mixed-precision CGNE, and the Krylov solvers on $\tilde{A}_L$ without the multigrid (Table 9): iterations (top) and seconds per right-hand side (bottom) against the quark mass, $24^3 \times 48$. CGNE and multigrid: means over the four configurations, bars from the smallest to the largest value; the solvers without the multigrid: configuration 1. "Solve + setup/12": the setup spread over the 12 columns of one point-source propagator.

GCR(2) needs 9–10 on all four. The stall follows the small smeared lattice of Sec. V.D.4; the topological charge does not predict it. On the same right-hand side the dual cycle does not converge on 12^4 in setup S.

What did not work. We tried several other ways to solve with $D^\dagger$ from one setup. None is as good as $\Gamma_5 \mathcal{M} \Gamma_5$.

- *The exact map.* Eq. (2.13) gives the null vectors of $D^\dagger$ as $(D_-^\dagger)^{-1}\Gamma_5 v$, and $D^\dagger w = r$ as a solve with D followed by one with $D_-^\dagger$. Both need D_-^{-1}, which does not converge (Sec. IV.D).
- *Other prolongations for the dual cycle.* $\mathcal{P}$ from $y \cup \Gamma_5 y$ converges (8–11 iterations) but doubles $\mathcal{P}$ and quadruples the coarse operators. $\mathcal{P}$ from the left singular vectors $\tilde{A}_R y$, with or without a bootstrap, or from a second setup of $\tilde{A}_R^\dagger$, needs 12–300 iterations and up to a full setup more.
- *Other coarse solvers in the dual cycle.* Its stagnation comes from the Pauli–Villars factor, solved by BiCGStab: with GCR(4) there it converges, but each cycle is three times as expensive. MINRES on the Γ_5-Hermitian form does not help, since that form is wrong at order c.

TABLE 7. Iterations and solve seconds at $m = 0.001$ against the volume, one random right-hand side: ranges over the four configurations; multigrid: plus its mean setup time. BiCGStab: configuration 1. The 209 iterations on 12^4 (S) are a slow setup of Sec. V.D.4.

			12^4	16^4	24^4	$24^3 \times 48$
S	CGNE, mixed	it	234–289	358–781	1715–2657	3622–4936
		s	0.2	0.6–1.2	12.5–19.4	51.9–76.1
	BiCGStab on $\tilde{A}_L$	it	235	398	1905	3247
		s	0.4	2.0	49.9	169.5
	multigrid	it	12–209	12–14	15–20	23–53
		s	0.5–7.8 (+7)	0.6–0.8 (+18)	2.9–4.4 (+96)	10.3–26.0 (+209)
B	CGNE, mixed	it	505–584	818–1249	3388–4116	4232–4959
		s	0.3–0.4	1.3–1.9	24.6–29.8	61.1–70.9
	BiCGStab on $\tilde{A}_L$	it	368	663	1909	2085
		s	0.7	3.4	50.0	108.8
	multigrid	it	13–14	14–16	22–23	22–26
		s	0.4–0.5 (+6)	0.8–0.9 (+18)	5.2–5.5 (+102)	10.3–12.9 (+213)

TABLE 8. GCR iterations of a solve with $D^\dagger$ to $|r|^2/|b|^2 < 10^{-10}$ from one setup of D (32 test vectors, blocks 4^4). "rand": random right-hand side; "res": the residual after one sweep of $D^\dagger D$. Medians over the four configurations, in brackets the number that do not converge in 300 iterations ("stall": three or four; with two, the range of the other two). One inner cycle per step. For comparison, the solve with D takes 7–9 (16^4) and 7–12 (12^4) iterations on configuration 1.

	16^4, $m = 0.01$				12^4, $m = 0.001$			
	S		B		S		B	
method	rand	res	rand	res	rand	res	rand	res
dual cycle, GCR(2)	16	stall	9	13	stall	stall	13	17
dual cycle, GCR(8)	10	20 [1]	9	12	stall	stall	11	16
$\Gamma_5\mathcal{M}\Gamma_5$, GCR(2)	9	10	11	11	12–13 [2]	stall	12	14
$\Gamma_5\mathcal{M}\Gamma_5$, GCR(8)	9	10	10	11	11	20	12	12

What these solves are for. In HMC each force evaluation needs one solve with D and one with $D^\dagger$, and with $\Gamma_5\mathcal{M}\Gamma_5$ one setup serves both. The solves with $D^\dagger$ were measured on 12^4 and 16^4 only, and the cost of the multigrid in HMC is left to future work (Sec. VII).

C. What each part contributes

The four parts of this subsection take the method apart: what the coarse correction adds to the operator A', what the lift $\hat{t}$ adds to the basis, what the Σ-Hermitian Pauli–Villars solve costs at $c \neq 0$, and how much the parameters matter.

1. *Krylov on $\tilde{A}_L$ without the multigrid*

The comparison with CGNE mixes two effects: the operator A', on which a Krylov method works without the normal equation, and the coarse correction. Table 9 separates them with BiCGStab on $\tilde{A}_L$ alone and GCR(2) on $\tilde{A}_L$ preconditioned by the Chebyshev smoother alone, both in double precision (mixed precision exists only for CG on the normal equation) and with the smoother parameters of Table 3, not tuned for this use. BiCGStab needs about as many iterations as double-precision CGNE (3247 against

TABLE 9. Solvers on $24^3 \times 48$, one random right-hand side, iterations and solve seconds (multigrid: plus its setup).

	setup S		setup B	
	$m = 0.01$	$m = 0.001$	$m = 0.01$	$m = 0.001$
CGNE, double	1166, 31.5	3602, 97.4	1447, 39.1	4228, 114.3
CGNE, mixed	1166, 16.9	4059, 58.0	1448, 20.9	4232, 60.4
BiCGStab on $\tilde{A}_L$	747, 39.1	3247, 169.5	672, 35.1	2085, 108.8
GCR(2), Chebyshev	236, 51.4	1789, 389.6	139, 44.1	430, 136.3
multigrid	15, 5.8 (+198)	28, 12.9 (+200)	15, 6.8 (+205)	26, 12.3 (+208)

TABLE 10. GCR(2) iterations on $\tilde{A}_L$ with $\mathcal{P}$ made of different lifts of the vectors y, $m = 0.001$, one random right-hand side; the other parameters as in Table 3; setup seconds on $24^3 \times 48$. 12^4 and 16^4: medians over the four configurations, in brackets the number that do not converge in 2000 iterations ("stag.": three or four); $24^3 \times 48$: configuration 1.

			12^4		16^4		$24^3 \times 48$		setup (s)
lifts	y	columns	S	B	S	B	S	B	S, B
$\{s, \hat{t}\}$	16	32	14	14	13	15	28	26	196, 211
$\{s, Ds\}$	16	32	stag.	30	stag.	22	148	30	193, 207
$\{s\}$	16	16	stag.	31	stag.	22	stag.	32	143, 153
$\{s\}$	32	32	90 [1]	32	27	15	28	13	346, 337

3602 at $m = 0.001$ in setup S), but one of its iterations applies $\tilde{A}_L$ twice and $\tilde{A}_L$ contains two Schur complements, so it costs about twice a CGNE iteration and takes 1.7 times its time. The polynomial cuts the iterations by two to five but raises the time. So A' alone does not beat CGNE, and the speed-up of the multigrid comes from its coarse correction.

2. *Choice of the prolongation basis*

The exactness (3.8) needs both lifts s and $\hat{t}$ of each y in $\operatorname{range}\mathcal{P}$, but the left form used in practice is not exact (Sec. III.A.3). To see what $\hat{t}$ does there, we replace it by Ds, which vanishes on the odd sites, or drop it (Table 10). With $\{s\}$ alone $\mathcal{P}$ has half the columns; to compare at the same size we also take 32 vectors y, which doubles the vectors of the filter and of the bootstrap.

With the same vectors y, the two lifts give the fewest iterations on every volume. In setup S, replacing $\hat{t}$ by Ds or dropping it stagnates on three of the four 12^4 and 16^4 configurations, and on $24^3 \times 48$ it costs 5 times the iterations (148 against 28) or a stagnation; in setup B it costs 1.2 to 2.2 times the iterations. So $\hat{t}$ helps also where the correction is not exact.

Twice as many vectors y with $\{s\}$ alone take the place of $\hat{t}$ only on the larger lattices. On 12^4 they do not (155 against 14 in setup S), which suggests that the Pauli–Villars solve needs the direction of $\hat{t}$ in $\operatorname{range}\mathcal{P}$; on 16^4 they come within two iterations, and on $24^3 \times 48$ they match the two lifts in setup S (28) and need half their iterations in setup B (13 against 26). They cost a setup 1.6–1.8 times as long and twice the host memory (38.7 against 19.7 GB). Estimated from one random right-hand side, a point-source propagator (12 columns) is faster with the two lifts in both setups (S: 351 against 506 s; B: 359 against 414 s); in setup B the 32 vectors y alone recover their longer setup after about 21 right-hand sides.

3. Γ_5-*Hermiticity for* $c \neq 0$

The loss of Γ_5-Hermiticity at $c \neq 0$ does not hurt the method. For $c = 0$ the coarse operators are Σ-Hermitian to rounding. For $c \neq 0$ they are not: by a few percent in random directions and up to 13% in the directions of the smallest singular values, growing linearly with c (Table 11), as Eq. (2.10) says. Yet

TABLE 11. $\delta(v) = \|\Sigma X \Sigma v - X^\dagger v\| / \|X^\dagger v\|$ for the coarse operators $X = \mathcal{D}_\mathrm{c} = \mathcal{P}^\dagger D \mathcal{P}$ and $\mathcal{P}^\dagger D_\mathrm{PV} \mathcal{P}$, $X^\dagger$ the exact adjoint, Σ the coarse Γ_5; $b - c = 1$ throughout (as usual for Möbius fermions), $L_s = 8$, $m = 0.001$ ($4^3 \times 8$: $m = 0.05$, 2^4 aggregates, 32 coarse components; 12^4: 4^4, 64). Random v (four) and the right singular vectors of the four smallest singular values; on 12^4 ranges over the four configurations (mean of the four random v per configuration). *Lanczos on $X^\dagger X$ not converged in 2000 restarts on some or all configurations (the low end of $\mathcal{D}_\mathrm{c}$ is crowded): approximate vectors.

	$\mathcal{P}^\dagger D \mathcal{P}$		$\mathcal{P}^\dagger D_\mathrm{PV} \mathcal{P}$	
	random	low	random	low
$4^3 \times 8$, $M_5 = 1$, $(b, c) = (1, 0)$	3.8×10^{-16}	9×10^{-16}	4.0×10^{-16}	9×10^{-16}
$4^3 \times 8$, $M_5 = 1$, $(b, c) = (1.5, 0.5)$	0.041	0.065–0.074	0.050	0.090–0.095
12^4 S, $(b, c) = (1.25, 0.25)$	0.012–0.014	0.035–0.054*	0.030–0.033	0.023–0.037
12^4 S, $(b, c) = (1.5, 0.5)$	0.023–0.027	0.065–0.110*	0.053–0.065	0.040–0.076
12^4 B, $(b, c) = (1.5, 0.5)$	0.031–0.032	0.113–0.132*	0.056–0.057	0.052–0.067

TABLE 12. GCR(2) iterations (coarse applications) with the Pauli–Villars factor solved by MINRES on $\mathcal{H}_\mathrm{c}$, which assumes $\delta = 0$, or by BiCGStab on the exact adjoint, both to 10^{-2}; 12^4, $m = 0.001$, $\tilde{A}_L$, otherwise Table 3. Medians over the four configurations, in brackets the number that do not converge in 2000 iterations.

	$c = 0.5$, $b = 1.5$		$c = 0$, $b = 1$	
	MINRES	BiCGStab	MINRES	BiCGStab
setup S	14 (3308)	628 [1] (383020)	13 (2665)	25 (7329)
setup B	14 (2140)	19 (3262)	15 (1602)	29 (3078)

with MINRES on $\mathcal{H}_\mathrm{c}$ for the Pauli–Villars factor, the outer iterations at $c = 0.5$ are those of the Shamir operator ($c = 0$, a different operator, for which MINRES is exact). MINRES is also better than BiCGStab on the exact adjoint in every case (Table 12), and with BiCGStab the stout setup needs 84–883 iterations or does not converge. So the non-Hermitian part of $\mathcal{H}_\mathrm{c}$ is small enough for MINRES stopped at 10^{-2}. The same weakness of BiCGStab on this factor makes the dual cycle of Sec. V.B.2 stagnate.

4. Sensitivity to the parameters

The method is robust within a window of each parameter and fails outside it (Table 13). The smoother interval must suit the setup, the bootstrap is necessary, and more accurate coarse solves do not help: on 12^4 with GCR(2) the most accurate ones stagnate in setup S (Sec. V.D.4). Larger setups (smaller aggregates, more test vectors) reduce the iteration count, but on 12^4 not the time (Table 14). All runs of the table use GCR(2), and its stagnations may be fixed points of the kind of Sec. V.D.4.

Fewer iterations are not always less time (Table 14): 2^4 aggregates need 8–9 iterations but a larger coarse lattice, so the solve and the setup take two to three times as long as with 4^4. Likewise, 48 test vectors save a quarter of the iterations and lose in setup and in time per iteration. In the other direction, 16 test vectors (a coarse dimension of 32) halve the setup; the solve is as fast in setup B, while in setup S one configuration stagnates. More smoother steps ($\nu = 12$, 16 for setup S) need fewer iterations (12, 9) at the same setup and solve time. On $24^3 \times 48$, however, 16 test vectors do not suffice (setup S, $m = 0.001$, point source): they halve the setup there too (116 against 207 s), but the columns that converge need 910–940 iterations and the others stop at the cap of 2000; 24 (184 s of setup) need 162 iterations for the first column, against 29–32 with 32. With a random right-hand side the four configurations give the same picture: 16 test vectors need 230–337 iterations or do not converge in 2000, and 24 need 63–172, against 23–53 with 32 (setups of 114–127, 147–164 and 204–215 s).

TABLE 13. GCR(2) iterations when one parameter of Table 3 is changed, 12^4, $m = 0.001$, $\tilde{A}_L$, one random right-hand side: medians over the four configurations, in brackets the number of configurations that do not converge in 1000 iterations ("stag.": three or four). Base: 14 in both setups (Sec. V.A); configuration 4 has the slow setup of Sec. V.D.4. Setup times (medians, seconds) where they change much.

change	setup S	setup B
aggregates 2^4 / 3^4 / 6^4	8 (19 s) / 12 / 38 [1]	9 (12 s) / 11 / 20
test vectors 8 / 16 / 24 / 48	stag. / 26 [1] / 18 / 10 (16 s)	187 [1] / 19 / 16 / 11 (12 s)
smoother $\nu = 4$ / 6 / 8 / 12 / 16	30 / 16 / 14 / 12 / 9	stag. / 81 [1] / 26 / 14 / 13
interval $\kappa_s = 5$ / 10 / 30 / 100	18 / 14 / stag. / stag.	stag. / 27 / 14 / stag.
coarse solves 10^{-1} / 10^{-2} / 10^{-4} / 10^{-8}	25 [1] / 14 / 18 [1] / stag.	18 / 14 / 16 / 17
bootstrap rounds 0 / 1 / 2 / 3	stag. / 32 [1] / 14 / 18	228 / 40 [1] / 14 / 13
right form $\tilde{A}_R$	12	14

TABLE 14. The changes of Table 13 that move the cost, with timings: iterations / setup seconds / solve seconds, medians over the four configurations as in Table 13. 12^4, $m = 0.001$, $\tilde{A}_L$, GCR(2), one random right-hand side.

change	setup S	setup B
base	14 / 7 / 0.5	14 / 6.5 / 0.5
aggregates 2^4	8 / 19 / 1.4	9 / 12 / 1.2
aggregates 3^4	12 / 8 / 0.5	11 / 7 / 0.4
aggregates 6^4	38 [1] / 7 / 1.1	20 / 7 / 0.5
test vectors 16	26 [1] / 4 / 0.6	19 / 4 / 0.5
test vectors 24	18 / 5.5 / 0.5	16 / 5.5 / 0.4
test vectors 48	10 / 16 / 0.7	11 / 12 / 0.8
smoother $\nu = 4$	30 / 8 / 1.2	stag.
smoother $\nu = 6$	16 / 7 / 0.6	81 [1] / 6 / 2.6
smoother $\nu = 12$	12 / 7.5 / 0.6	(base)
smoother $\nu = 16$	9 / 8 / 0.5	13 / 7 / 0.5
bootstrap rounds 1	32 [1] / 4 / 1.1	40 [1] / 4 / 0.9
bootstrap rounds 3	18 / 9.5 / 0.7	13 / 9 / 0.5

For Wilson fermions, Whyte, Stathopoulos and Romero [56] obtain the effect of more test vectors at the coarse size of fewer: they generate more, truncate them by a singular value decomposition on each aggregate and chirality (as $\mathcal{P}$ is split here by Γ_5), and stream the decomposition to save device memory. Applied here, the truncated basis would no longer contain s and $\hat{t}$ exactly, so the exactness (3.8) would be lost; whether the gain outweighs that loss is open.

D. Why the method converges

1. *Field of values: the coarse correction amplifies*

The four parts of this subsection ask why the method converges, as it does in Sec. V.B.1, although the usual theory does not apply. Two notions from non-Hermitian linear algebra are used. The *field of values* $W(X) = \{z^\dagger X z / z^\dagger z\}$ contains the spectrum, and its leftmost point is the smallest eigenvalue of the Hermitian part $(X + X^\dagger)/2$; GCR is guaranteed to converge at every restart length when W lies in the right half-plane [44]. The *ε-pseudospectrum* is the set of z where $\sigma_{\min}(z - X) < \varepsilon$, i.e. where $\|(z - X)^{-1}\| > 1/\varepsilon$; for a normal X it is the ε-neighborhood of the spectrum, for a non-normal one it

TABLE 15. Real part of the field of values $W(X)$ and $\|I - X\|_2$ for $X = \tilde{A}_R\mathcal{S}$, $\tilde{A}_R E$, $\tilde{A}_R\mathcal{M}$ (exact coarse solves). $m = 0.05$ on $4^3 \times 8$ (2^4 aggregates, 8 filtered low modes), $m = 0.001$ otherwise, with the multigrid of the mass scan (16 low modes, 32 test vectors, 4^4, Chebyshev of Table 3) but no levels below the coarse lattice. On 24^4 only $W(\tilde{A}_R)$ is given (Krylov dimensions 10 and 12 agree to six digits); there the field of values of $\tilde{A}_R\mathcal{M}$ did not settle (the two dimensions differ by 2%) and the eigensolvers did not fit in 24 GB of device memory.

	$\mathrm{Re}\, W(\tilde{A}_R)$	$\|I - \tilde{A}_R\mathcal{S}\|$	$\|I - \tilde{A}_R E\|$	$\mathrm{Re}\, W(\tilde{A}_R\mathcal{M})$	$\|I - \tilde{A}_R\mathcal{M}\|$
$4^3 \times 8$, $M_5 = 1$	$[-5.0 \times 10^{-3}, 2.26]$		18.5	$[0.772, 1.222]$	
12^4 S	$[-4.59 \times 10^{-3}, 1.98]$	1.018	1.94×10^4	$[-23.9, 30.8]$	54.8
12^4 B	$[-5.02 \times 10^{-2}, 4.46]$	1.112	2.86×10^3	$[-1.49, 4.17]$	5.64
16^4 S			1.54×10^4	$[-15.6, 27.1]$	
16^4 B			2.5×10^3	$[-1.42, 4.27]$	
24^4 S	$[-1.31 \times 10^{-2}, 1.98]$				
24^4 B	$[-5.08 \times 10^{-2}, 4.48]$				

can be much larger, and it bounds the convergence of full GMRES [51].

The condition number of an eigenvalue, $\kappa = 1/|y^\dagger x|$ for its right and left eigenvectors x and y normalized to $\|x\| = \|y\| = 1$, is 1 for a normal matrix; a large κ means that the eigenvalue moves by $\kappa\varepsilon$ under a perturbation of size ε, which widens the pseudospectrum around it (Secs. V.D.2, V.D.3).

The coarse correction amplifies some vectors by 10^3–10^4, and the field of values of the preconditioned operator reaches into the left half-plane (this subsection), so the bound of Eisenstat, Elman and Schultz [44] for GCR is lost. Secs. V.D.2, V.D.3 and V.D.4 give what remains: the eigenvalues, the pseudospectra and the truncated GCR used in practice.

The Hermitian part of $\tilde{A}_R\mathcal{M}$ was computed with $\mathcal{M}$ made a fixed linear map: both coarse factors solved to $|r|^2/|b|^2 < 10^{-20}$ in double precision with the exact adjoints, and $\mathcal{M}^\dagger$ built with the same $\mathcal{P}$ and the same smoother interval, so that $\langle z|\tilde{A}_R\mathcal{M}w\rangle$ and $\langle\mathcal{M}^\dagger\tilde{A}_R^\dagger z|w\rangle$ agree to 10^{-10}. The cycle was also taken apart: the smoother alone, $\mathcal{S} = p(\tilde{A}_R)$, the coarse correction alone, E of Eq. (3.2), and the cycle $\mathcal{M}$ (Table 15).

The coarse correction is what pushes the field of values into the left half-plane. On $4^3 \times 8$ the field of values of $\tilde{A}_R\mathcal{M}$ lies in the right half-plane, whereas on 12^4 and 16^4 it does not. Table 15 shows where the change comes from.

- The field of values of $\tilde{A}_R$ itself hardly depends on the volume. From 12^4 to 24^4 its upper edge moves by less than 1%, and its lower edge goes from -4.6×10^{-3} to -1.3×10^{-2} (S) and stays at -5×10^{-2} (B).
- The smoother is harmless: $\|I - \tilde{A}_R\mathcal{S}\| \approx 1$.
- The coarse correction amplifies: $\|I - \tilde{A}_R E\|$ is 10^3–10^4, against 18.5 on $4^3 \times 8$. $\tilde{A}_R E$ is the identity on the vectors the correction is built from (Eq. (3.8)), but it amplifies other directions by three to four orders of magnitude, like a strongly oblique projection.

Eick and Mawhinney [54] describe the same mechanism for a renormalization-group coarse lattice. The coarse inverse is large on the low modes of the coarse operator. The interpolated correction is not exactly a low mode of the fine operator, and the part that misses comes back amplified. In their case the fine operator is $D^\dagger D$, whose spectrum reaches 520 against 0.05 for its lowest modes; it supplies the four orders of magnitude, and they need a Chebyshev filter of degree 150–300. In our case $\tilde{A}_R$ is close to the identity on the high modes, because the Pauli–Villars operator divides them out, and its field of values ends at 2 (S) and 4.5 (B). The large factor is the coarse inverse itself. The post-smoothing, a Chebyshev

TABLE 16. Eigenvalues of $\tilde{A}\mathcal{M}$ (exact coarse solves) by Krylov–Schur: the smallest real part, the condition numbers $\kappa = 1/|y^\dagger x|$ of the six eigenvalues of smallest real part (left eigenvectors from $\mathcal{M}^\dagger \tilde{A}^\dagger$), the eigenvalues farthest from 1 with their κ, and the radius about 1 of the disc that holds all the others. $m = 0.001$. On 16^4 the smallest real part and κ are ranges over the four configurations, the outliers and the radius are those of configuration 1.

	$\min \mathrm{Re}\,\lambda$	κ (6 lowest)	outliers (κ)	others: $\lvert\lambda - 1\rvert \le$
$4^3 \times 8$, $m = 0.05$, R	0.857	2–4		
$4^3 \times 8$, $m = 0.05$, L	0.854	2.8		0.156 (Ritz)
12^4 S, R	0.868	6.3–14.5	$5.86 + 0.86i$ (11.2)	0.434
12^4 S, L	0.861	7.2–26.2	$5.40 + 0.89i$ (13.9)	0.455
12^4 B, R	0.831	9.4–20.3		0.753
12^4 B, L	0.830			0.747
16^4 S, R	0.700–0.794	3.6–22.8	$10.63 + 0.19i$ (4.38)	0.474
			$3.13 + 0.02i$ (4.74)	
16^4 B, R	0.698–0.783	9.5–42.9		0.840

polynomial of degree 7–11 on the upper part of the spectrum, acts as the filter, and the whole cycle keeps $\|I - \tilde{A}_R\mathcal{M}\|$ at 55 (S) and 5.6 (B).

So the GCR bound of Ref. [44] does not apply. The field of values is also not what decides the iteration count. If we replace E by the Galerkin correction of $\tilde{A}_R$ itself, $\mathcal{P}(\mathcal{P}^\dagger \tilde{A}_R \mathcal{P})^{-1}\mathcal{P}^\dagger$ (built densely), the field of values of setup B moves into the right half-plane, $[0.121, 1.99]$, and $\|I - \tilde{A}_R\mathcal{M}\|$ drops to 1.84 (setup S: $[-3.62, 13.1]$, 16.8). Yet GCR(8) to 10^{-16} then needs more iterations: 16 against 13 (B) and 12 against 10 (S).

2. Spectrum: far from the origin

The spectrum, unlike the field of values, stays far from the origin. Every eigenvalue lies at $\mathrm{Re}\,\lambda \ge 0.70$, with moderate condition numbers (Table 16). On configuration 1 setup S has one (12^4) or two (16^4) isolated outliers of large modulus, and all other eigenvalues lie within 0.43–0.47 of 1; on configurations 2–4 of 16^4 the eigenvalue farthest from 1 lies at $|\lambda| = 2.7$–4.6. The left and right forms have the same spectrum to the precision shown. At $m = 0.005$ on 16^4 the lowest real parts are 0.802 (S) and 0.790 (B).

3. Pseudospectra and a bound for GMRES

For a non-normal operator the eigenvalues alone do not bound the convergence of GMRES; the ε-pseudospectrum, the set where $\sigma_{\min}(z - X) < \varepsilon$, does [35, 51]. If a contour Γ of length L encloses the spectrum of X but not the origin and $\sigma_{\min}(z - X) \ge \varepsilon$ on it, then for any polynomial p_k with $p_k(0) = 1$,

$$\frac{\|r_k\|}{\|r_0\|} \le \|p_k(X)\| \le \frac{L}{2\pi\varepsilon} \max_{z\in\Gamma} |p_k(z)|. \tag{5.1}$$

An isolated outlier λ_o outside Γ is removed by the factor $(1 - z/\lambda_o)$ of p_k: then $p_k(X) = p_k(X)(I - P_o)$ with P_o the spectral projector, $\|(z - X)^{-1}(I - P_o)\| \le \kappa_o/\varepsilon$, and with a circle $|z - c| = r$ and $p_k(z) = (1 - z/\lambda_o)(1 - z/c)^{k-1}$,

$$\frac{\|r_k\|}{\|r_0\|} \le \kappa_o \frac{r}{\varepsilon} \frac{|\lambda_o - c| + r}{|\lambda_o|} \Big(\frac{r}{c}\Big)^{k-1}. \tag{5.2}$$

On 16^4 setup S has two outliers; the factor κ_o is then replaced by $\|I - P_1 - P_2\| \le 1 + \kappa_1 + \kappa_2$ and there are two factors of the form $(|\lambda_o - c| + r)/|\lambda_o|$. On the real axis between the origin and the spectrum, $\sigma_{\min}(z - \tilde{A}_R\mathcal{M}) \ge 0.024$ (S) and ≥ 0.031 (B) on 12^4 ($z = -0.5 \ldots 0.7$), and 0.107 (S), 0.271 (B) at the origin.

TABLE 17. $\min_\Gamma \sigma_{\min}(z - \tilde{A}_R\mathcal{M})$ on circles $|z-c| = r$ (16 points each, or the number given), and the bound (5.1) or (5.2): the factor per iteration r/c, the constant, and the iterations it guarantees for $\|r_k\|/\|r_0\| < 10^{-9}$. For the dense circles ε is the lower bound $\min_i \sigma_{\min}(z_i) - h/2$ on the whole circle, h the spacing of the points ($\sigma_{\min}$ is 1-Lipschitz in z); otherwise the smallest sampled value. Lanczos to 10^{-3} relative in σ^2.

	c	r	$\min \sigma_{\min}$	r/c	constant	k for 10^{-9}
12^4 S	1	0.6	0.0285	0.6	220	53
12^4 S, 160 points	1	0.6	0.0281, ≥ 0.0164	0.6	380	54
12^4 S	1	0.5	0.0107	0.5	480	40
12^4 B	1.4	0.93	0.0985	0.66	9.4	57
12^4 B, 96 points	1.4	0.93	0.0980, ≥ 0.0676	0.66	13.8	58
12^4 B	1.3	0.87	0.0860	0.67	10.1	58
16^4 S	1	0.6	0.0262	0.6	190	53
16^4 S	1	0.55	0.0135	0.55	340	47
16^4 B	1.4	1.05	0.1295	0.75	8.1	80

TABLE 18. GCR(k) iterations to $|r|^2/|b|^2 < 10^{-18}$, 12^4, $m = 0.001$, the multigrid of Table 3 (no levels below the coarse lattice), one random right-hand side: medians over the four configurations, in brackets the number that do not converge in 1000 iterations; the point source: range over its 12 columns and the configurations. *Configuration 4 (the slow setup of Table 19): 97–521, one column not converged.

	coarse solves	$k = 2$	8	16	32	64
S, $\tilde{A}_L$	10^{-2}	14			12	
S, $\tilde{A}_R$	10^{-2}	12			11	
B, $\tilde{A}_L$	10^{-2}	14			13	
B, $\tilde{A}_R$	10^{-2}	15			14	
S, $\tilde{A}_L$, point source	10^{-2}	12–19*			12–15	
S, $\tilde{A}_L$	10^{-8}	no conv.	24 [1]	17	17	17
S, $\tilde{A}_L$	10^{-4}	18 [1]			16	

The bound guarantees convergence at a fixed rate, r/c per iteration, with no stagnation. It is pessimistic by a factor of 3–5 in iterations: GCR needs 10–17 with exact coarse solves (Sec. V.D.4). Its rate, however, is the one we measure on $24^3 \times 48$ on configurations 1 and 3. There the median factor per GCR step of $|r|^2$ is 0.23–0.34 (Table 21), that is 0.48–0.58 in $|r|$, against $r/c = 0.5$–0.6 for setup S. Configuration 2 in setup S converges more slowly, at a steady 0.56 per step in $|r|^2$. Setup B does not get worse from 12^4 to 16^4 ($\min \sigma$ grows from 0.098 to 0.13). In setup S the outliers of configuration 1 grow, but their condition numbers fall and ε stays the same.

What is not yet rigorous. On the two dense circles of 12^4 the spacing of the points is below ε, and since $\sigma_{\min}(z - X)$ is 1-Lipschitz in z the bound holds on the whole circle, $\sigma_{\min} \geq 0.0164$ (S) and ≥ 0.0676 (B), with 54 and 58 guaranteed iterations. The circles of 16^4 have only 16 points. The Lanczos estimates approach $\sigma^2_{\min}$ from above, the disc rests on the eigenvalues farthest from c found by Arnoldi, and the bound is for full GMRES with a fixed $\mathcal{M}$ (Secs. V.D.4, VII).

4. Truncated GCR: inexact coarse solves

Truncated GCR does not need accurate coarse solves, and with them it can fail. With the usual coarse solves to 10^{-2}, GCR(2) needs one to five iterations more than the full method on 12^4 (Table 18). With

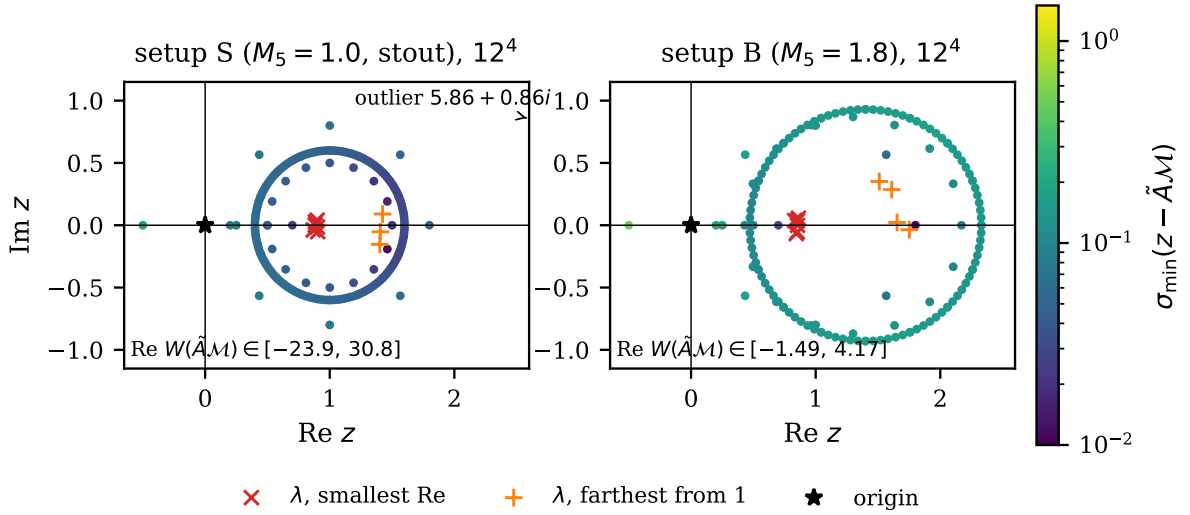


FIG. 5. 12^4, $m = 0.001$, exact coarse solves: the six eigenvalues of $\tilde{A}_R\mathcal{M}$ of smallest real part, the four farthest from 1, and $\sigma_{\min}(z - \tilde{A}_R\mathcal{M})$ (color) at the points where it was computed: the real axis, the contours of Table 17 and the dense circles $|z - 1| = 0.6$ (S) and $|z - 1.4| = 0.93$ (B). The field of values reaches far into the left half-plane (its real range in the corner), the spectrum does not.

accurate coarse solves, where $\mathcal{M}$ is close to the fixed map of Secs. V.D.1–V.D.3, GCR(2) stagnates there in setup S on all four configurations ($|r|^2 \approx 3 \times 10^{-8}$ after 1000 iterations on configuration 1), while GCR(8) converges in 17–26 on three of them and GCR(≥ 16) in 15–17 on all four. The pseudospectra guarantee the full method; a truncated method has no guarantee when the field of values contains the origin [44, 52]. On 16^4 and 24^4 GCR(2) converges with the same accurate solves (Table 20).

Inexact coarse solves do not keep GCR(2) converging on 12^4 by leaving out the directions in which E amplifies. Take the vector x at the left end of the field of values of $\tilde{A}_R\mathcal{M}$ (exact coarse solves, 12^4). With coarse solves to 10^{-2} the cycle gives $\mathrm{Re}\, x^\dagger \tilde{A}_R\mathcal{M}x = -25.2$ (S) and -1.43 (B), against -24.3 and -1.49 with exact ones, and $\|(I - \tilde{A}_R E)x\| = 1.16 \times 10^3$ and 1.03×10^2, against 1.13×10^3 and 1.12×10^2. With tolerances 10^{-8} and 10^{-4} they agree with the exact solves to within 1%. What the tolerance changes is that the cycle is no longer a fixed linear map: with Krylov coarse solves stopped early, it depends on its input.

How GCR(2) stagnates. The stagnation is a fixed point of GCR(2) with a fixed preconditioner. On 12^4 in setup S, with three right-hand sides, $|r|^2$ falls for about 15 steps and then does not change in any digit for the next 480, at 2×10^{-11}, 6×10^{-8} and 5×10^{-10}. Meanwhile $|\langle r|\tilde{A}_L\mathcal{M}r\rangle|/(\|r\|\,\|\tilde{A}_L\mathcal{M}r\|)$ falls from about 0.8 to 10^{-10}–10^{-9}. The new direction is then orthogonal to the residual, and since $\mathcal{M}$ is fixed the next step repeats it. A field of values that contains the origin admits such a residual. Any variation of the cycle leaves the fixed point: on configuration 1 of 12^4 the exact cycle and two fixed perturbations of it stagnate in all 15 runs, and random perturbations, down to 2% of the cycle, converge in all 75 (Table 20). On the four quenched 16^4 configurations, on the dynamical one and on 24^4 the exact cycle does not reach a fixed point in the runs made. A fixed perturbation can still produce one there ($\delta\Gamma_5\mathcal{M}r$ on 24^4), so whether a fixed cycle stagnates depends on the cycle and on the lattice. Of the four 12^4 configurations, 1 and 4 reach it and 2 and 3 do not, so the fixed point depends on the configuration; which property of it decides is open. It is neither the topological charge (configurations 3 and 4 both have $Q = +2$, and 24^4 has $Q = +4$ and converges) nor the odd extent of the 3^4 coarse lattice (3^4 aggregates give a 4^4 coarse lattice and stagnate as well).

Dependence on the setup seed. On 12^4 with stout smearing about one setup in eight makes GCR(2) slow. Without smearing, in setup B, and on 16^4 none of the setups tested is; on $24^3 \times 48$ one configuration is

TABLE 19. Slow setups: setups with more than 30 iterations or no convergence in 2000 / setups, and the range of the others; $m = 0.001$, $L_s = 8$, $\tilde{A}_L$, one random right-hand side to $|r|^2/|b|^2 < 10^{-18}$. 12^4: configurations 1–4; 16^4: 1–3 and the dynamical one; $24^3 \times 48$: 3 and 4. Last column: peak device memory of the variant on $24^3 \times 48$ (setup S, configuration 2, seed 21).

lattice	setup	variant	seeds	slow	others	memory
12^4	S	GCR(2)	1–40	19/160	11–30	20.2 GB
	S	GCR(4)	1–10	3/40	12–20	22.1 GB
	S	GCR(8)	1–10	0/40	11–19	does not fit
	S	GCR(16)	1–10	0/40	11–14	does not fit
	S	48 test vectors	1–10	0/40	9–15	does not fit
	S	quark coarse 10^{-4}	1–40	2/160	11–30	20.2 GB
	S, no smearing	GCR(2)	1–10	0/40	13–15	
	B	GCR(2)	1–20	0/80	13–14	
16^4	S	GCR(2)	1–20	0/60	12–15	
	S, dynamical	GCR(2)	1–20	0/20	11	
$24^3 \times 48$	S	GCR(2)	1–5	0/10	18–28	

slow with every seed. The setup filters random starting vectors; their seed (21 in all other runs) changes the coarse space but not the configuration, the operator or the right-hand side, and the solution agrees to the stopping tolerance. Table 19 repeats the solve at $m = 0.001$ with many seeds. On 12^4 it uses four configurations of Table 23. In setup S, 19 of 160 setups need more than 30 iterations and 13 of them more than 200 or do not converge in 2000. The rate per configuration, 7, 6, 4 and 2 of 40 (configurations 1, 3, 2, 4), does not distinguish the configurations at these counts. The same configurations with $M_5 = 1$ and no smearing give 0 of 40, and setup B gives 0 of 80. On 16^4 setup S gives 0 of 60 on three quenched configurations and 0 of 20 on the dynamical one, and on $24^3 \times 48$ seeds 1–5 need 18–28 iterations on configurations 3 and 4. Configuration 2 of $24^3 \times 48$ needs 53, 36 and 45 iterations with seeds 21, 7 and 99, against 18–28 on the other three; that excess is shared by all three seeds and so belongs to the configuration.

A slow setup gives a cycle of the usual quality, and GCR(2) fails to use it. With the same cycles GCR(16) needs 11–14 iterations in all 40 setups tested. In the slow runs $|r|^2$ falls normally for a few steps and then stays nearly constant, for up to 300 steps at $|r|^2/|b|^2 \approx 7 \times 10^{-10}$ on configuration 1 with seed 2, before it falls again; this is the behavior near a fixed point of Table 20, reached here with inexact coarse solves. The slowness belongs to the setup: five of the six slow setups of the first ten seeds are slow for the point source as well (44 to more than 500 iterations). At $L_s = 12$ and 16 the same occurs, in 7 of 20 setups on configurations 2 and 4. Why smearing on the small lattice produces these setups is open.

GCR(8) and 48 test vectors remove the slow setups on 12^4 but do not fit $24^3 \times 48$ in 24 GB: each kept direction costs two fine vectors (1 GB there; GCR(4) peaks at 22.1 against 20.2 GB), and the setup with 48 test vectors runs out of device memory. Coarse solves for the quark factor to 10^{-4} cost no memory and leave 2 of 160 setups slow. On $24^3 \times 48$ they make each iteration 22–40% more expensive: configuration 2 drops to 43, 33 and 33 iterations at about the same solve time, and the other three configurations take 9–34% longer. A slow setup shows itself in the first solve as a plateau of $|r|^2$; the setup can then be repeated with another seed on the same configuration, at the cost of one more setup. We have not automated this.

On $24^3 \times 48$ no run shows a plateau (Table 21, Fig. 6); the first step reduces $|r|^2$ by about 10^{-5} in every run.

TABLE 20. GCR(2) on $\tilde{A}_L$ in setup S, $m = 0.001$, $\alpha = 0.6$, to $|r|^2/|b|^2 < 10^{-16}$ (at most 500 iterations, 300 on 24^4): converged runs / runs, with their iterations. "exact": coarse solves to 10^{-20} in double precision, BiCGStab on the exact Pauli–Villars adjoint; "fixed": the exact cycle plus $\delta\,\Gamma_5\mathcal{M}r$ or plus a fixed multiple of the smoother polynomial, δ the relative change that coarse solves to 10^{-2} make (0.04–0.23); "random": the exact cycle plus a random vector of relative size $\delta = 0.02$–0.5, drawn anew at each application; "solver": the cycle of the solver (MINRES, 10^{-2}), whose correction did not fit next to the exact one on 24^4 (OOM). Q: the topological charge. Aggregates 4^4 unless stated; configurations 2–4: three right-hand sides.

lattice, Q	exact	fixed $\Gamma_5\mathcal{M}$	fixed $p(\tilde{A}_L)$	random	solver
12^4 config. 1, $+1$	0/5	0/5	0/5	75/75 (34–120)	5/5 (12–14)
12^4 config. 2, $+1$	3/3 (21–24)				3/3 (12–13)
12^4 config. 3, $+2$	3/3 (18–20)				3/3 (10)
12^4 config. 4, $+2$	0/3				3/3 (84–94)
12^4, 3^4 aggregates	0/3				3/3 (11–12)
12^4, 2^4 aggregates	2/3 (20, 23)				3/3 (7)
16^4 config. 1, 0	5/5 (33–36)	5/5	5/5	75/75 (31–114)	5/5 (12)
16^4 config. 2, $+1$	3/3 (24–25)				3/3 (12–13)
16^4 config. 3, $+1$	3/3 (29–31)				3/3 (11)
16^4 config. 4, $+3$	3/3 (25–29)				3/3 (12)
16^4 dynamical, 0	5/5 (13)	5/5	5/5	20/20 (13–20)	5/5 (10)
24^4, $+4$	2/2 (78, 85)	0/2	2/2 (41, 42)	4/4 (84–205)	OOM

What this establishes. As usual for multigrid preconditioners of lattice Dirac operators, the convergence of the truncated flexible outer method is established numerically. GCR(2) with coarse solves stopped at 10^{-2} converged without plateaus in every run of this section on 16^4 and $24^3 \times 48$, mostly at the rate the pseudospectral bound of Sec. V.D.3 gives for the full method; on 12^4 with smearing about one setup in eight reaches a plateau (Table 19). No guarantee is available for a truncated method whose preconditioner varies from step to step. Accurate coarse solves do not make the solver faster. On 16^4 the exact cycle takes 24–36 iterations on the four configurations, as many as BiCGStab on the exact adjoint stopped at 10^{-2} (37–47 on configuration 1). The solver's cycle takes 11–13. It differs from the exact one in two ways, the Σ-adjoint $\Sigma\mathcal{D}_c^{\mathrm{PV}}\Sigma$ in place of $(\mathcal{D}_c^{\mathrm{PV}})^\dagger$ and MINRES stopped at 10^{-2}. The first is not the whole difference: at $c = 0$, where the two adjoints coincide, MINRES still needs 13 iterations against 25 with BiCGStab on 12^4 (medians, Table 12). The cycle of the solver converged in every run of Table 20.

TABLE 21. Residual history of GCR(2), $24^3 \times 48$, $\tilde{A}_L$, the multigrid of Table 3: steps, median and largest factor of $|r|^2$ per step, and the longest stretch of steps with a factor above $1/2$. Configuration 1 unless stated; the steps differ from those of the mass scan by at most one (Sec. V.A).

setup	m, α	steps	median	worst	slow stretch
S	0.001, 0.6	28	0.315	0.564	2 steps
S	0.001, 1.0	27	0.338	0.479	none
S	0.002, 0.6	23	0.233	0.426	none
S	0.002, 1.0	23	0.253	0.354	none
B	0.001, 0.6	26	0.299	0.357	none
B	0.001, 1.0	28	0.317	0.403	none
B	0.002, 0.6	23	0.246	0.300	none
B	0.002, 1.0	25	0.280	0.347	none
configurations 2 and 3, $m = 0.001$, $\alpha = 0.6$					
S, config. 2		53	0.559	0.731	9 steps
S, config. 3		23	0.244	0.371	none
B, config. 2		25	0.271	0.307	none
B, config. 3		22	0.239	0.294	none

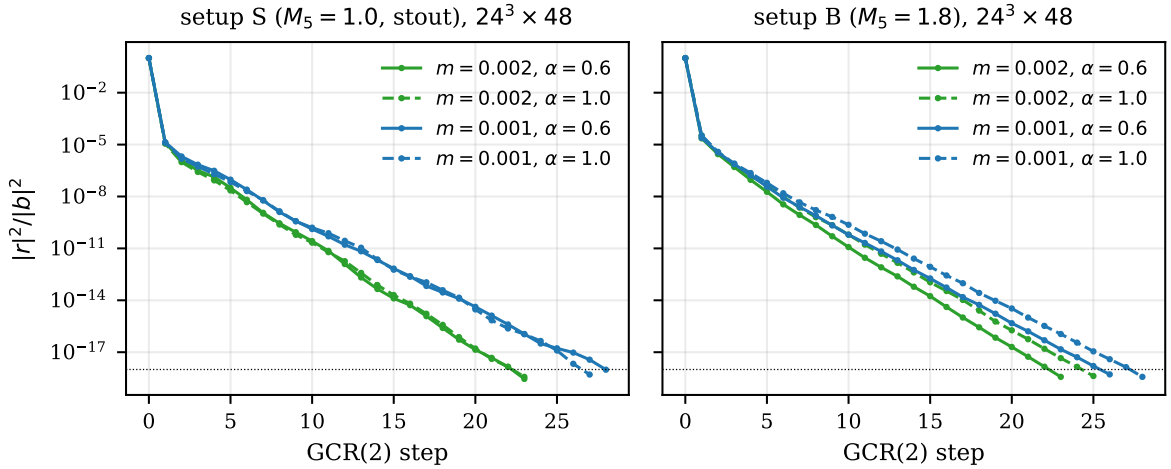


FIG. 6. The GCR(2) residual histories of Table 21, $24^3 \times 48$.

VI. Summary

Boyle and Yamaguchi named three reasons why the Pauli–Villars multigrid did not beat red-black CG in four-dimensional QCD: it worked on the whole lattice, its coarse space kept the fifth dimension, and its non-normal coarse solves were not deflated. The present method addresses them in turn. It solves the even-odd system. Its aggregates contain the whole fifth dimension, so the correction solves two nearest-neighbor systems on a four-dimensional coarse lattice. Its Pauli–Villars coarse system is solved in a Σ-Hermitian form, and levels below the coarse lattice take the place of deflation. With the same test vectors, the lift $\hat{t}$ that makes the correction exact for the right form avoids stagnation or cuts the iterations by up to a factor of 5 in setup S (1.2–2.2 in setup B) also in the left form, which is not exact; one lift of twice as many vectors matches it on $24^3 \times 48$ at 1.6–1.8 times the setup time.

The multigrid removes most of the critical slowing down. On quenched configurations on one GPU the multigrid iterations on four $24^3 \times 48$ configurations grow by a factor of 1.6–4.1 from $m = 0.1$ to $m = 0.001$, against 18–27 for mixed-precision even-odd CGNE. At $m = 0.001$ a solve is faster than with mixed-precision CGNE from 24^4 on, by 2.9–6.0 on $24^3 \times 48$, and a point-source propagator takes half the time, setup included. Without the coarse correction, both outer GCR(2) with the Chebyshev smoother alone (across the mass scan on $24^3 \times 48$) and BiCGStab on $\tilde{A}_L$ alone (across all volumes and masses) are slower than mixed-precision CGNE, confirming that the acceleration comes from the multigrid correction rather than merely from the lower condition number of the unsquared operator $\tilde{A}_L$. One setup serves Neff's α. The field of values of the preconditioned operator reaches into the left half-plane; its spectrum and pseudospectra account for the convergence.

The evidence is limited to four quenched configurations per volume (one for the field of values and the pseudospectra), volumes up to $24^3 \times 48$, and comparisons without other multigrid solvers in the same code. A comparison with HDCG and with the coarsening of $D_{\mathrm{PV}}^{\dagger} D$ in one implementation, larger volumes, and the use in HMC are the next steps (Sec. VII). For propagators, applying the coarse operators and the transfer to the 12 columns of a point source together would raise their intensity from about one flop per byte to about ten, as for Wilson-clover multigrid [42]. Because batching multiple right-hand sides substantially increases device memory usage, we will treat the batch size as an optional parameter to balance memory footprint and solve time. Finally, for the $D^{\dagger}$ solves required by HMC at $c \neq 0$, we compared several candidate constructions—including the dual cycle of $\tilde{A}_R^{\dagger}$, the exact map through D_{-}^{-1}, and enlarged or separate prolongations—and found that the cheapest option, wrapping the unchanged cycle of D in Γ_5 with no second setup or extra coarse operators, works best and costs only two to three iterations more than solving with D. Although HMC itself is not implemented here, this lays the groundwork for fast force evaluations and, together with tuning the multigrid parameters, will be the next stage of our study.

Code. All results were obtained with RutticeQCD, `https://github.com/atsanroger/RutticeQCD`, a rewrite of Bridge++ [33, 34] in Rust whose GPU backend uses the CUDA kernels of the Wilson and Möbius operators of Bridge++. Where Bridge++ computes the same quantity, the results of RutticeQCD are compared with it number by number: the plaquette, the Wilson, clover and domain-wall operators applied to a vector, solver iteration counts, meson correlators, and single HMC trajectories. The domain-wall multigrid, which Bridge++ does not have, is checked against CGNE, whose propagators it reproduces (Sec. V.B.1).

VII. Open problems

The following questions remain open. Those the measurements answer are discussed where they arise: the left form in Sec. III.A.3, $c \neq 0$ in Secs. III.B and V.C.3, the field of values in Secs. II.A and V.D.1–V.D.3.

1. **Parameters from the operator.** The smoother interval $[\lambda_{\mathrm{hi}}/\kappa_s, \lambda_{\mathrm{hi}}]$ uses a ratio chosen per setup,

and a wrong one makes the solve stagnate (Sec. V.C.4); the setup works with singular values of $\tilde{A}_R$, which for a non-normal operator need not match its eigenvalues. How to set the interval from the operator, whether an ellipse with complex foci should replace the real interval, and how the required number of test vectors scales with the volume remain to be understood. Local coherence suggests that the number of test vectors does not grow with the volume, yet 16 suffice on 12^4 and 24 do not on $24^3 \times 48$ (Sec. V.C.4). Part of this difference may come from the small physical box of 12^4 ($\approx (1.2\,\mathrm{fm})^4$, with fewer low modes) and from the additional coarse levels used on $24^3 \times 48$ (Table 3); the low modes may also vary in the fifth dimension more than vectors not split in s can follow (Eick and Mawhinney [54] find their fifth-dimensional profiles to differ between lattices); a test vector split in s, as it is now split in chirality, would keep s and $\hat{t}$ in the range of $\mathcal{P}$, and Lüscher's coherence test would measure what the coarse space captures.

2. **Convergence theory.** The pseudospectral bound rests on Lanczos estimates of $\sigma_{\min}$, which approach it from above, and on the eigenvalues farthest from the center found by Arnoldi (Sec. V.D.3). Obtaining a bound from an inclusion theorem, and tracking how the outliers of setup S and the pseudospectra develop beyond 16^4, remain to be done; on 24^4 the Arnoldi iteration does not fit in 24 GB of device memory. No theory covers truncated GCR with a preconditioner that varies from step to step. On 12^4 in setup S, GCR(2) with accurate coarse solves can reach a residual orthogonal to its new direction and stay there (Sec. V.D.4), and on the residual right-hand side of HMC the solve with $D^\dagger$ needs GCR(8) on all four configurations (Sec. V.B.2). GCR(2) does not reach such a fixed point on 16^4 and 24^4, although 24^4 has $Q = +4$; the solves with $D^\dagger$ were tested on 16^4 but not on 24^4. With the solver's inexact cycle the same fixed points appear for about one setup seed in eight on 12^4 with stout smearing, on every configuration tested, and in none of the setups tested without smearing, in setup B, on 16^4 (quenched and dynamical) and on three of the four $24^3 \times 48$ configurations (Sec. V.D.4, Table 19). Why smearing on the small lattice produces them is not understood. A safeguard that changes the direction when $|\langle r|\tilde{A}\mathcal{M}r\rangle|/(\|r\|\,\|\tilde{A}\mathcal{M}r\|)$ falls below a threshold would remove such fixed points for one inner product per step; we have not implemented it. A quantitative explanation for why $N = c[Y, W]D^{-1}$ has a small square, beyond the heuristic of Sec. IV, is also lacking.

3. **Formulations and coarse spaces.** The comparisons here are with CGNE and with Krylov solvers on $\tilde{A}_L$ alone. How the two nine-point factors compare with the 81-point HDCG operator and with the 33-point coarsening of $D^\dagger_{\mathrm{PV}} D$ [43], in cost per iteration and in iterations, needs all of them in one code on the same lattices.

Acknowledgments

I thank Issaku Kanamori and Hideo Matsufuru for their kind permission to port Bridge++ to Rust, without which this study would not have been possible; the structure of RutticeQCD follows Bridge++ closely, and its correctness rests on a number-by-number comparison with it. I am also grateful to my colleagues at my current company for teaching me the practical details of algebraic multigrid in computational fluid dynamics, which inspired the two-lift prolongation $\{s, \hat{t}\}$. Although computational fluid dynamics and lattice QCD lie far apart in physics, their numerical methods have much to learn from one another. Finally, I thank my friend Sakura Itatani for helping me find a reference on local coherence.

TABLE 22. The notation of this work and that of other references. Equation numbers are those of the references; "the same" means the same operator.

	This work	Ref. [17]	Ref. [45]	Bridge++ code
Height	$M_5 > 0$	$-M_5$, $M_5 = O(-1)$ (2.13)	M_0	M_0, `domain_wall_height`
Fifth dim.	$s = 0, \ldots, L_s - 1$	$s = 1, \ldots, L_s$	$s = 1, \ldots, N_s$	$0, \ldots, N_s - 1$, `extent_of_5th_dimension`
Wilson part	$W = (4 - M_5) - \frac{1}{2}K$	$D^{\mathrm{Wilson}}(M_5)$ (2.13)	$D_{\mathrm{W}}(-M_0)$	$\frac{1}{2\kappa}D_{\mathrm{W}}$, $\kappa = 1/(8 - 2M_0)$
$D_\pm$	$D_+ = 1 + bW$, $D_- = cW - 1$ (Eq. (2.3))	$D_+ = bD^{\mathrm{W}} + 1$, $D_- = cD^{\mathrm{W}} - 1$ (2.18)	the same (2)	—
D	Eq. (2.7)	$D_{\mathrm{DW}}(m)$ (2.20)	$D(m)$ (1)	`Fopr_Domainwall`, mode `D`
C	$d_+ + d_- Y$, Eq. (2.14)	—	$\mathrm{diag}(D_{ee}, D_{oo})$ (4)	D_{ee}, D_{oo} of `_eo`; `m_dp` $= \alpha d_+$, `m_dm` $= \alpha d_-$
Neff's α	DQ, Eq. (2.21)	— (its $\alpha = \frac{b+c}{b-c}$)	—	`parameter_alpha`: DQ, site part CQ
Γ_5	$\gamma_5 R$	$\gamma_5 R D_-^{-1}$ (2.22)	—	—
4d quark	$P_-\psi_0 + P_+\psi_{L_s-1}$	$P_-\Psi_1$, $P_+\Psi_{L_s}$	—	the same

Appendices

A. The notation of other references

Table 22 relates Eqs. (2.6)–(2.21) to Brower, Neff and Orginos [17], to Kanamori, Chen and Matsufuru [45] and to the code of Bridge++ [34] (`Fopr_Domainwall`, `Fopr_Domainwall_eo`, with Neff's α as `parameter_alpha`). Eq. (2.4) is the tridiagonal matrix of Ref. [17], Eq. (2.20), and of Ref. [45], Eq. (1), with $D_+ = 1 + bW$ on the diagonal, $D_- P_+$ below and $D_- P_-$ above it, $-mD_- P_\pm$ in the corners, and s shifted by one. The operators D, C and $D^{(\alpha)}$ are thus the same in all four; the differences are names and signs:

- $D_\pm$ are those of Refs. [17, 45], $D_- = cW - 1$. The hop Y of Eq. (2.6) carries a minus sign, so $D = D_+ - D_- Y$ and $d_- = 1 - c(4 - M_5)$ is minus the site part of D_-.
- The height $M_5 > 0$ here is M_0 in Ref. [45] and in Bridge++ (`domain_wall_height`); Ref. [17] writes M_5 for the Wilson mass, $M_5^{\mathrm{BNO}} = -M_5$, Eq. (2.13). The caption of Fig. 1 of Ref. [45] gives $M_0 = -1$ for the height 1 of its text.
- The α of Ref. [17], $(b + c)/(b - c)$, is the scale factor of the Möbius kernel (2 at $b = 1.5$, $c = 0.5$), not Neff's α of Eq. (2.21).
- $\Gamma_5 = \gamma_5 R D_-^{-1}$ in Ref. [17], Eq. (2.22); here $\Gamma_5 = \gamma_5 R$.

$D^{(\alpha)}$ here and the operator of Bridge++ with `parameter_alpha` were compared on a 12^4 configuration with stout links, $L_s = 8$, $m = 0.01$: at $\alpha = 0.6$ and $\alpha = 1$ CG on the even-odd normal equation took the same number of iterations for every column of a point source, and the pseudoscalar correlators agree to 10^{-11}. Bridge++ returns $\psi^{(\alpha)}$; $\psi = Q\psi^{(\alpha)}$.

B. Spectra with other preconditioners

Figure 7 reproduces Fig. 3 of Ref. [45]. It shows the spectra that Sec. II.A compares with that of A': the unpreconditioned D, the site-diagonal DC^{-1}, the Pauli–Villars form $D_{\mathrm{PV}}^\dagger D$ of Ref. [41], and its block-diagonal approximations.

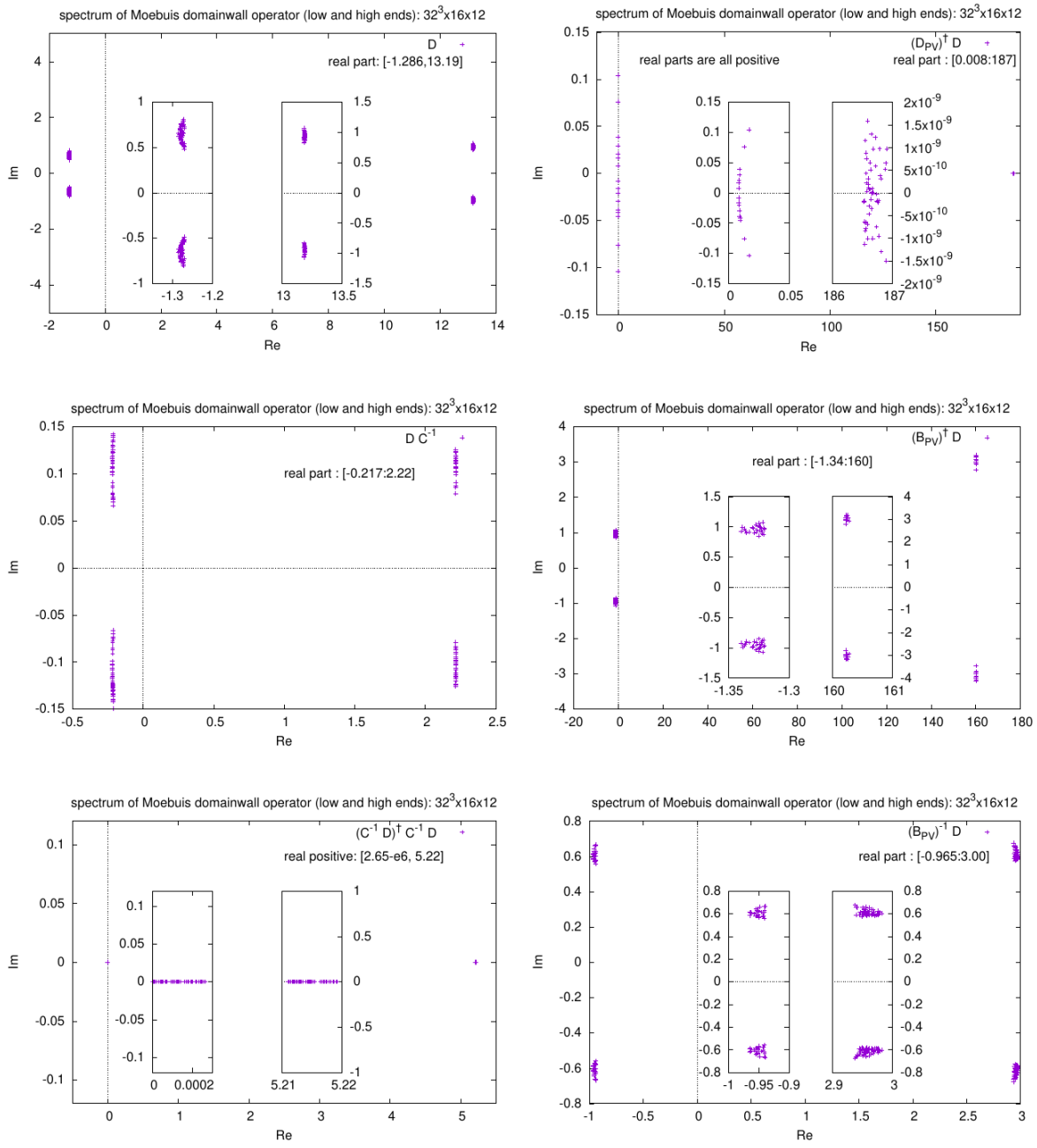


FIG. 7. Low and high ends of the spectrum of the Möbius operator on the configuration of Fig. 2, with preconditioners. Left column: D, DC^{-1}, $(C^{-1}D)^\dagger C^{-1}D$. Right column: $D^\dagger_{\rm PV}D$, $B^\dagger_{\rm PV}D$, $B^{-1}_{\rm PV}D$, where $B_{\rm PV}$ is the block-diagonal approximation of $D_{\rm PV}$ (inverted approximately in the last panel). From Ref. [45], Fig. 3.

TABLE 23. The gauge configurations: plaquette, t_0/a^2, and the topological charge at t_0. The logs and the generation histories are listed with the data.

lattice		plaquette	t_0/a^2	Q_{clover}	Q_{imp}	Q
12^4	1	0.594250	3.19	$+0.94$	$+1.00$	$+1$
	2	0.593201	3.29	$+0.95$	$+1.00$	$+1$
	3	0.594204	3.78	$+1.78$	$+1.96$	$+2$
	4	0.594103	4.04	$+1.88$	$+1.99$	$+2$
16^4	1	0.593509	3.54	$+0.05$	$+0.01$	0
	2	0.593847	3.55	$+0.95$	$+1.00$	$+1$
	3	0.594395	3.18	$+0.96$	$+1.00$	$+1$
	4	0.593685	3.03	$+2.53$	$+2.88$	$+3$
16^4 dynamical		0.593809	6.90	-0.003	-0.0002	0
24^4	1	0.593524	3.65	$+3.63$	$+3.92$	$+4$
	2	0.593789	3.41	$+1.77$	$+1.94$	$+2$
	3	0.593297	3.16	-3.55	-3.93	-4
	4	0.593786	3.24	-4.69	-5.05	-5
$24^3 \times 48$	1	0.593548	3.20	$+4.47$	$+4.78$	$+5$
	2	0.593630	3.18	$+7.83$	$+8.42$	$+8$
	3	0.593689	3.19	$+3.46$	$+3.53$	3 or 4
	4	0.593395	3.23	-1.57	-1.87	-2

C. Gauge configurations

All quenched configurations use the plaquette gauge action at $\beta = 6.0$ and were generated by HMC from a cold start, 400 trajectories of length 0.5 (on $24^3 \times 48$, 400 from a cold start and then 200); these are configurations 1. Configurations 2–4 continue the same chains with trajectories of length 0.5 and are taken 100 trajectories apart, after 502, 602 and 702 trajectories (702, 802 and 902 on $24^3 \times 48$). The dynamical 16^4 configuration was generated from the quenched one by HMC in setup S with mixed-precision CG: 60 trajectories at $m = 0.05$, then 40 at $m = 0.01$, of length 1 with Omelyan steps $[20, 2]$. Table 23 gives the plaquette, the scale t_0 ($t^2E = 0.3$, E of the clover field strength) and the topological charge at t_0 after the Wilson flow (step 0.01), of the clover field strength and improved with rectangles ($c_{\text{plaq}} = 5/3$, $c_{\text{rect}} = -1/12$). On the larger lattices the two definitions differ by up to 1 at t_0.

D. Krylov solvers

The multigrid uses three Krylov solvers, flexible GCR for the outer system, MINRES and BiCGStab for the coarse factors and FGMRES in the bootstrap, and the comparison uses CG. The algorithms below are those of RutticeQCD, which runs the same code on the CPU and on the device; ϵ is the tolerance on $|r|^2/|b|^2$.

Flexible GCR(k). The outer solver of Sec. III.E is GCR [44] with a preconditioner $\mathcal{M}$ that may change from one step to the next, keeping the last k directions (Algorithm 4). Each step applies $\mathcal{M}$ once, makes the new $q = \tilde{A}z$ orthogonal to the q of the kept directions, and minimizes the residual along it; the step length $\langle q|r\rangle$ vanishes when q is orthogonal to r, which is the fixed point of Sec. V.D.4. After $n_{\max}$ steps the true residual is recomputed and the solve restarts.

Algorithm 4. Flexible GCR(k) for $\tilde{A}x = b$ with the preconditioner $\mathcal{M}$.

repeat
$r \leftarrow b - \tilde{A}x$
for at most $n_{\max}$ steps, while $|r|^2 \geq \epsilon|b|^2$ **do**
$z \leftarrow \mathcal{M}r; \quad q \leftarrow \tilde{A}z$
for each of the last k pairs (z_i, q_i) **do**
$\gamma \leftarrow \langle q_i|q\rangle; \quad q \leftarrow q - \gamma q_i; \quad z \leftarrow z - \gamma z_i$
end for
$z \leftarrow z/|q|; \quad q \leftarrow q/|q|; \quad$ keep (z, q)
$\beta \leftarrow \langle q|r\rangle; \quad x \leftarrow x + \beta z; \quad r \leftarrow r - \beta q$
end for
until $|b - \tilde{A}x|^2 < \epsilon|b|^2$

MINRES. The Pauli–Villars factor is solved by MINRES [22] on the Σ-Hermitian form $\mathcal{H}_{\rm c}$ (Sec. III.B), in the form of Ref. [53], Algorithm 2.4: Lanczos in the inner product of the preconditioner, which must be Hermitian positive definite and fixed ($|H_{xx}|^{-1}$ or the additive levels of Eq. (3.12)), with Givens rotations (Algorithm 5). As a coarse solve it starts from $x = 0$ and stops on the residual estimate η of the recurrence, measured in the norm of the preconditioner.

Algorithm 5. Preconditioned MINRES for $Hx = g$, H Hermitian, $\mathcal{M}_H$ Hermitian positive definite.

$v \leftarrow g - Hx; \quad v_- \leftarrow 0; \quad z \leftarrow \mathcal{M}_H v; \quad \gamma \leftarrow \langle z|v\rangle^{1/2}; \quad \gamma_- \leftarrow 1; \quad \eta \leftarrow \gamma$
$c, c_- \leftarrow 1; \quad s, s_- \leftarrow 0; \quad w, w_- \leftarrow 0$
while $\eta^2 \geq \epsilon\,\langle \mathcal{M}_H g|g\rangle$ **do**
$z \leftarrow z/\gamma; \quad \delta \leftarrow \mathrm{Re}\langle Hz|z\rangle$
$v' \leftarrow Hz - (\delta/\gamma)v - (\gamma/\gamma_-)v_-; \quad z' \leftarrow \mathcal{M}_H v'; \quad \gamma' \leftarrow \mathrm{Re}\langle z'|v'\rangle^{1/2}$
$a_0 \leftarrow c\delta - c_- s\gamma; \quad a_1 \leftarrow (a_0^2 + \gamma'^2)^{1/2}; \quad a_2 \leftarrow s\delta + c_- c\gamma; \quad a_3 \leftarrow s_-\gamma$
$c' \leftarrow a_0/a_1; \quad s' \leftarrow \gamma'/a_1; \quad (w_-, w) \leftarrow \big(w,\, (z - a_3 w_- - a_2 w)/a_1\big)$
$x \leftarrow x + c'\eta\, w; \quad \eta \leftarrow -s'\eta$
$(v_-, v, z) \leftarrow (v, v', z'); \quad (\gamma_-, \gamma) \leftarrow (\gamma, \gamma'); \quad (c_-, c, s_-, s) \leftarrow (c, c', s, s')$
end while

FGMRES. Each bootstrap round of Sec. III.C replaces a vector y_j by a few steps of flexible GMRES [23] on $\tilde{A}_R x = y_j$ from $x = 0$, with the cycle as a right preconditioner (Algorithm 6). The basis is orthogonalized by classical Gram–Schmidt applied twice, since one pass loses orthogonality on the strongly non-normal $\tilde{A}_R\mathcal{M}$, and the least-squares problem of the Hessenberg matrix is updated by Givens rotations at every step.

Algorithm 6. FGMRES(n) for $Ax = b$ with a right preconditioner $\mathcal{M}$.

repeat
$r \leftarrow b - Ax$; $\beta \leftarrow |r|$; $v_0 \leftarrow r/\beta$
for $k = 0, \dots, n-1$, while the residual estimate is $\geq \epsilon^{1/2}|b|$ **do**
$z_k \leftarrow \mathcal{M} v_k$; $w \leftarrow A z_k$
for two passes **do**
for $i = 0, \dots, k$ **do** $h_{ik} \leftarrow h_{ik} + \langle v_i | w \rangle$; $w \leftarrow w - \langle v_i | w \rangle v_i$
end for
end for
$h_{k+1,k} \leftarrow |w|$; $v_{k+1} \leftarrow w/h_{k+1,k}$; update the Givens rotations and the residual estimate
end for
$c \leftarrow \arg\min_c |\beta e_0 - Hc|$; $x \leftarrow x + \sum_i c_i z_i$
until $|b - Ax|^2 < \epsilon |b|^2$

CGNE. The reference solver of Sec. V.B.1 is CG on $\mathcal{K}^\dagger \mathcal{K}$ with $\mathcal{K} = C_e^{-1} S$, the even-odd operator of Bridge++, with Neff's α as for the multigrid, in double or in mixed precision.

For $D\psi = \eta$ the even-odd reduction of Sec. II.C gives $\mathcal{K}^{(\alpha)} y = C_e^{-1} \eta'_e$ with $\mathcal{K}^{(\alpha)} = Q_e^{-1} \mathcal{K} Q_e$, Eq. (2.27), and CG solves its normal equation $Ay = b$, $A = \mathcal{K}^{(\alpha)\dagger} \mathcal{K}^{(\alpha)}$, $b = \mathcal{K}^{(\alpha)\dagger} C_e^{-1} \eta'_e$; ψ_o and $\psi = Q \psi^{(\alpha)}$ follow as for the multigrid. The stopping criterion is $|b - Ay|^2/|b|^2 < 10^{-18}$. Since the unknown is y itself and A has $\mathcal{K}^{(\alpha)}$ on the right, Bridge++ and Ref. [50] call this method CGNR; we keep the name CGNE that is common in lattice QCD.

Mixed precision. The mixed-precision solver runs the CG recurrence in single precision and corrects it by reliable updates [25, 26] in double precision (Algorithm 7): each time the single-precision $|\hat{r}|^2$ has fallen by a factor $\delta = 10^{-2}$ below its largest value since the last update, the single-precision solution is added to y and the true residual $b - Ay$ is recomputed in double precision; CG then goes on with the same search direction and stops only on the double-precision residual. With $\delta = 0$ the algorithm is plain CG in double precision. On $24^3 \times 48$ the mixed-precision solver needs the iterations of double precision within 1% except at $m \leq 0.002$ in setup S, where it needs up to 30% more (14% on configuration 1), and takes 0.54–0.61 times the time, with the same final residuals and 7.2 GB of device memory against 6.9 GB in double precision.

Algorithm 7. CGNE with reliable updates. Hats: single precision; $\hat{A}$ is A on the single-precision links.

$r \leftarrow b - Ay$; **if** $|r|^2 < \epsilon|b|^2$ **return** y

$\hat{r} \leftarrow r$; $\hat{p} \leftarrow \hat{r}$; $\hat{x} \leftarrow 0$; $\rho \leftarrow |\hat{r}|^2$; $\rho_{\max} \leftarrow \rho$

loop

$\alpha \leftarrow \rho / \langle \hat{p}|\hat{A}\hat{p}\rangle$; $\hat{x} \leftarrow \hat{x} + \alpha\hat{p}$; $\hat{r} \leftarrow \hat{r} - \alpha\hat{A}\hat{p}$; $\rho' \leftarrow |\hat{r}|^2$

if $\rho' < \delta\rho_{\max}$ **or** $\rho' < \epsilon|b|^2$ **then** ▷ reliable update

$y \leftarrow y + \hat{x}$; $\hat{x} \leftarrow 0$; $r \leftarrow b - Ay$

if $|r|^2 < \epsilon|b|^2$ **return** y

$\hat{r} \leftarrow r$; $\rho' \leftarrow |\hat{r}|^2$; $\rho_{\max} \leftarrow \rho'$

else

$\rho_{\max} \leftarrow \max(\rho_{\max}, \rho')$

end if

$\hat{p} \leftarrow \hat{r} + (\rho'/\rho)\,\hat{p}$; $\rho \leftarrow \rho'$

end loop

BiCGStab. The coarse solves of the quark factor (Sec. III.B) use BiCGStab [21], Algorithm 8, with a right preconditioner $\mathcal{M}_A$. It bounds the angle between t and s from below as in Ref. [24], and after a breakdown or $n_{\max}$ steps it restarts from the true residual. The same algorithm, with $\mathcal{M}_A = 1$ and in double precision, is the Krylov solver on $\tilde{A}_L$ alone of Sec. V.C.1.

Algorithm 8. BiCGStab for $Ax = g$ with a right preconditioner $\mathcal{M}_A$; $\omega_c = 0.7$.

repeat

$r \leftarrow g - Ax$; $r_0 \leftarrow r$; $p \leftarrow r$; $\rho \leftarrow \langle r_0|r\rangle$

for at most $n_{\max}$ steps, while $|r|^2 \geq \epsilon|g|^2$ **do**

$\hat{p} \leftarrow \mathcal{M}_A p$; $v \leftarrow A\hat{p}$; $\alpha \leftarrow \rho/\langle r_0|v\rangle$; $s \leftarrow r - \alpha v$

$\hat{s} \leftarrow \mathcal{M}_A s$; $t \leftarrow A\hat{s}$; $\omega \leftarrow \langle t|s\rangle/|t|^2$

$\gamma \leftarrow |\langle t|s\rangle|/(|t|\,|s|)$; **if** $\gamma < \omega_c$ **then** $\omega \leftarrow \omega\,\omega_c/\gamma$

$x \leftarrow x + \alpha\hat{p} + \omega\hat{s}$; $r \leftarrow s - \omega t$; $\rho' \leftarrow \langle r_0|r\rangle$

$p \leftarrow r + (\rho'/\rho)(\alpha/\omega)\,(p - \omega v)$; $\rho \leftarrow \rho'$

end for

until $|g - Ax|^2 < \epsilon|g|^2$

Two even-odd forms. Boyle and Yamaguchi [40] used CG on $S^\dagger S$, which for Shamir fermions needed 3224 iterations against 3880 for the left-preconditioned form. Table 24 and Fig. 8 compare the two on the three quenched lattices at $m = 0.001$, in double precision with one random right-hand side, for $b - c = 1$ and five values of c; all other parameters are those of setups S and B.

The two operators differ by a site-local factor on the left, $S = C_e\mathcal{K}$, and with Neff's α, $SQ_e = C_eQ_e\mathcal{K}^{(\alpha)}$. Their condition numbers therefore differ at most by the condition number of that factor. Since

TABLE 24. CG iterations on $\mathcal{K}^\dagger\mathcal{K}$ / on $S^\dagger S$, $m = 0.001$, $b = 1 + c$, $\alpha = 1$ unless stated, one random right-hand side to $|r|^2/|b|^2 < 10^{-18}$, double precision. Final $|D\psi - \eta|^2/|\eta|^2 \le 4 \times 10^{-16}$ in every run except $(2, 1)$ on $24^3 \times 48$, S (up to 2×10^{-15}).

setup	lattice	$c = 0$	0.25	0.5	0.75	1
S	12^4	343/371	352/336	358/390	362/445	362/494
S	16^4	543/575	533/517	552/601	559/682	566/762
S	$24^3 \times 48$	4527/4979	5459/5222	5576/6061	5653/6869	5512/7414
B	12^4	622/514	657/579	664/681	653/763	633/827
B	16^4	1033/839	1039/917	1056/1083	1060/1238	975/1262
B	$24^3 \times 48$	2954/2439	6269/5532	5620/5770	3248/3772	1924/2443
S, $\alpha = 0.6$	12^4	234/364	226/331	234/344	249/378	263/415
S, $\alpha = 0.6$	16^4	373/576	346/511	358/528	378/576	403/638
S, $\alpha = 0.6$	$24^3 \times 48$	3095/4881	3506/5166	3602/5349	3790/5785	3878/6127
B, $\alpha = 0.6$	12^4	451/521	478/545	505/601	525/660	542/717
B, $\alpha = 0.6$	16^4	742/851	762/877	817/983	840/1068	827/1086
B, $\alpha = 0.6$	$24^3 \times 48$	2137/2474	4539/5194	4228/5097	2576/3239	1620/2093

$C = d_+ + d_- Y$ with $\|Y\| = 1$ (Eq. (2.7)), the singular values of C lie in $[d_+ - |d_-|,\, d_+ + |d_-|]$ and

$$\kappa(C) \le \frac{d_+ + |d_-|}{d_+ - |d_-|}, \qquad d_+ = 1 + b(4 - M_5), \quad d_- = 1 - c(4 - M_5); \tag{D.1}$$

for $\alpha < 1$ the bound is $1/\alpha$ times larger, since $\kappa(Q) = 1/\alpha$. The measured ratios of the iterations stay within this range too, although the bound is on the condition numbers and not on the iteration counts.

Within the bound the order depends on c. In setup B the ratio of the iterations grows smoothly with c, from 0.81–0.83 for Shamir fermions to 1.27–1.31 at $(b, c) = (2, 1)$, and crosses 1 near the $(b, c) = (1.5, 0.5)$ used in the main text. For Shamir fermions $S^\dagger S$ is the faster form by 1.20–1.23, close to the $3880/3224 = 1.20$ of Ref. [40]. In setup S the ratio is above 1 except at $c = 0.25$, where $S^\dagger S$ needs 3–5% fewer iterations; we have no explanation for this dip. The three lattices agree to within 0.04 in every case, so the order is a property of the operator and its parameters, not of the configuration.

Neff's α favors $\mathcal{K}^\dagger\mathcal{K}$. With $\alpha = 0.6$, $S^\dagger S$ needs 46–58% more iterations in setup S and 14–32% more in setup B, for every c and on all three lattices, Shamir fermions included. In $\mathcal{K}^{(\alpha)}$ the factor $C_e Q_e$ is divided out on the left, whereas in SQ_e the rescaling acts on the right only. The singular values of the two operators would show whether this is the cause. For the setups of this work $\mathcal{K}^\dagger\mathcal{K}$ is the faster form, and the multigrid is compared with it.

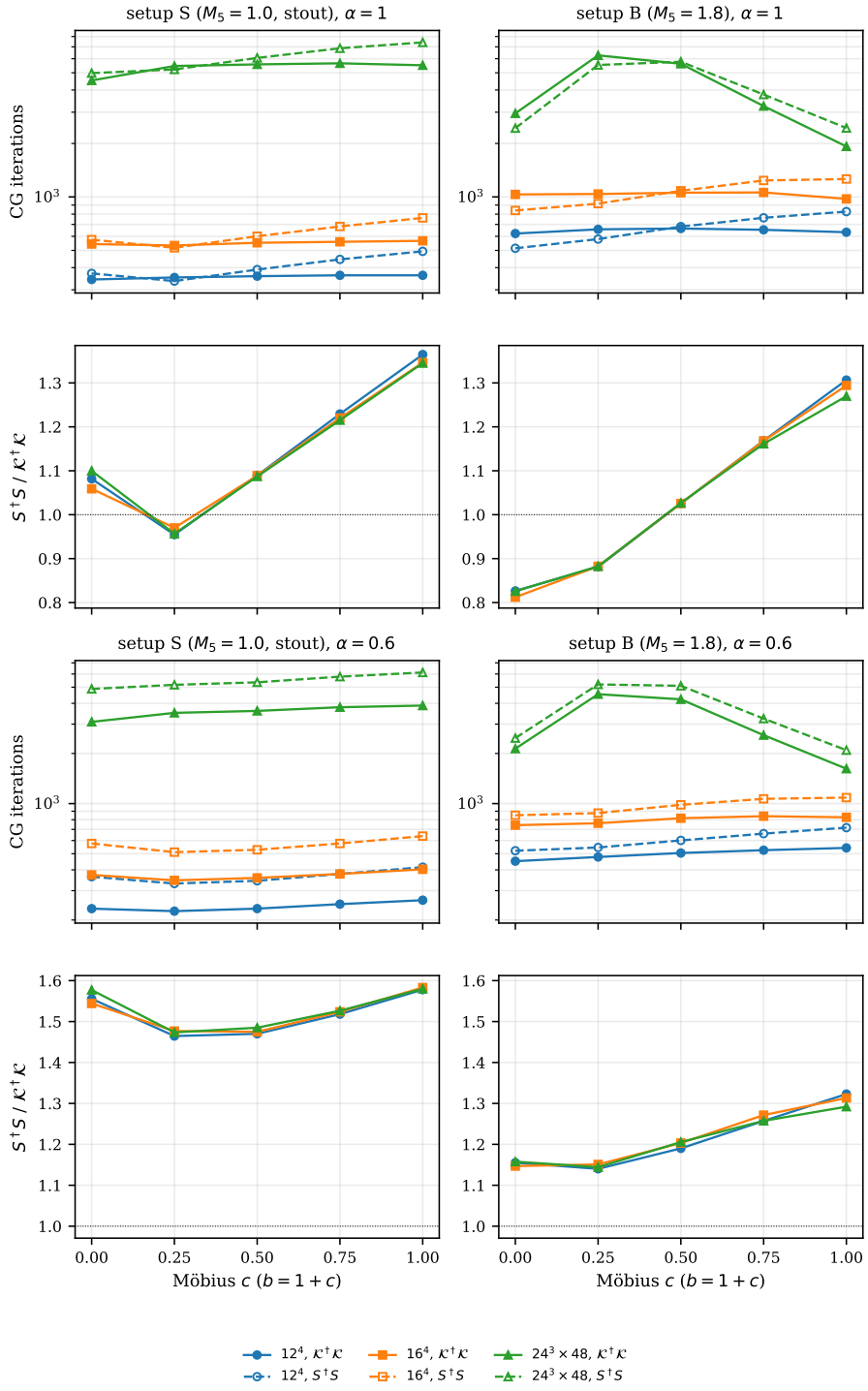


FIG. 8. CG iterations on $\mathcal{K}^\dagger\mathcal{K}$ (filled, solid) and on $S^\dagger S$ (open, dashed), with $\mathcal{K} = 1 - C_e^{-1} D_{eo} C_o^{-1} D_{oe}$ (the even-odd operator of Bridge++) and $S = C_e - D_{eo} C_o^{-1} D_{oe}$ (the Schur complement of Grid's default `SchurDiagMooeeOperator`, there on the odd sites), against the Möbius c, $b = 1 + c$, and below each their ratio; above 1 the left-preconditioned form is faster. Upper four panels $\alpha = 1$, lower four $\alpha = 0.6$. The data of Table 24.

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